

INSERTION AND EXPANSION OF CYLINDERS AROUND THE UNIT BALL

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ABSTRACT. A unit ball is surrounded with a ring of n equal infinite circular cylinders. Each cylinder touches the ball and its two neighbors. Can one more cylinder of the same radius be added if the configuration may be rearranged? Can the n cylinders instead be enlarged, and by how much? We prove that insertion is impossible for $3 \leq n \leq 7$ and possible for $n \geq 9$. The case $n = 8$ remains open. The case $n = 6$ is Kuperberg's problem from 1990, which was resolved recently. We prove that the ring of four cylinders cannot be expanded. This settles a conjecture of Ogievetsky and Shlosman from 2018. For $n \geq 5$ the largest possible expansion radius remains unknown.

1. INTRODUCTION

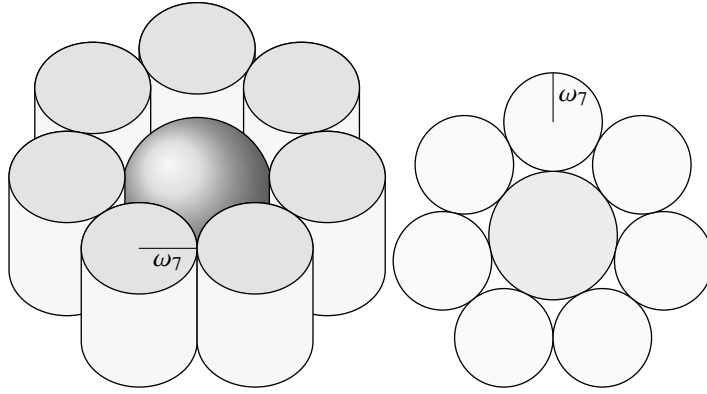


FIGURE 1. The ring of seven pairwise tangent cylinders around the unit ball (left) and its equatorial cross-section (right).

1.1. Formulation. A *cylinder* means a closed solid right circular cylinder of infinite length. Two cylinders are *non-overlapping* if their interiors are disjoint. A cylinder *touches* the unit ball B if it meets B but not the interior of B . For every integer $n \geq 3$ it is possible for n pairwise non-overlapping cylinders of radius

$$\omega_n = \frac{\sin \frac{\pi}{n}}{1 - \sin \frac{\pi}{n}} \quad (1)$$

to simultaneously touch the unit ball. The canonical way is the *ring configuration* of Figure 1. The cylinders are perpendicular to a common plane, and their axes pass through the vertices of a regular n -gon. Each of them is tangent to its two neighbors and to the ball. Much of the earlier work on cylinders touching a ball concerns the following two questions, for each fixed $n \geq 3$.

Question 1.1 (Insertion). Can one more cylinder of radius ω_n be added, for a total of $n + 1$? The cylinders may be rearranged, but they must be non-overlapping and touch the unit ball.

Question 1.2 (Expansion). Let $\omega' > \omega_n$. Can the n cylinders be enlarged to the common radius ω' and rearranged so that they are non-overlapping and touch the unit ball? For which ω' is the answer yes?

1.2. Cylinder-packing pairs. All results of this paper are statements about the following set of pairs.

Definition 1.3. $\mathcal{P}_{\text{cyl}}$ is the set of pairs $(n, \omega) \in \mathbb{N} \times (0, \infty)$ such that no more than n pairwise non-overlapping cylinders of radius ω can simultaneously touch the unit ball. We call these *cylinder-packing pairs*.

The following lemma is elementary and is also a consequence of Lemma 2.6 in Section 2.

Lemma 1.4. *If $(n, \omega) \in \mathcal{P}_{\text{cyl}}$, then $(n', \omega') \in \mathcal{P}_{\text{cyl}}$ for every integer $n' \geq n$ and every $\omega' \geq \omega$.*

In the ring, adjacent axes are at distance $2(1 + \omega_n) \sin \frac{\pi}{n} = 2\omega_n$. So the ring consists of n non-overlapping cylinders of radius ω_n . With Lemma 1.4 this gives

$$(n-1, \omega) \notin \mathcal{P}_{\text{cyl}} \quad \text{for every } 0 < \omega \leq \omega_n \quad (n \geq 3). \quad (2)$$

In the pair language, insertion at n is possible if and only if $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$. Expansion to the radius ω' is possible if and only if $(n-1, \omega') \notin \mathcal{P}_{\text{cyl}}$. Kuperberg's question is whether $(6, 1) \in \mathcal{P}_{\text{cyl}}$.

Two quantities are read off from $\mathcal{P}_{\text{cyl}}$. For $\omega > 0$ let $\mathcal{N}(\omega)$ be the smallest n with $(n, \omega) \in \mathcal{P}_{\text{cyl}}$. It is the largest number of non-overlapping cylinders of radius ω that can touch the unit ball. For $n \geq 2$ let R_n be the largest ω with $(n-1, \omega) \notin \mathcal{P}_{\text{cyl}}$. It is the largest radius at which n non-overlapping cylinders can touch the unit ball. The maximum exists by Corollary 2.8. Question 1.2 asks for R_n , and (2) gives $R_n \geq \omega_n$.

1.3. Results.

Theorem 1.5 (No insertion for $3 \leq n \leq 7$). *For every $n \in \{3, 4, 5, 6, 7\}$ we have $(n, \omega_n) \in \mathcal{P}_{\text{cyl}}$. Hence insertion is impossible for these n .*

The case $n = 6$ is Kuperberg's problem, solved in [16, 15]. Theorem 1.5 places it in a uniform family.

Theorem 1.6 (Insertion for $n \geq 9$). *For every $n \geq 9$ we have $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$. Hence insertion is possible for these n .*

Theorem 1.7 (Ogievetsky–Shlosman conjecture). *For every $\omega > 1 + \sqrt{2}$, $(3, \omega) \in \mathcal{P}_{\text{cyl}}$. Equivalently, $R_4 = 1 + \sqrt{2}$.*

Theorem 1.7 proves the conjecture of Ogievetsky and Shlosman [18, 21, 17]. We also find all configurations of four non-overlapping cylinders of radius $1 + \sqrt{2}$ touching the unit ball. Theorem 3.4 shows that they form a single one-parameter family. The ring of five cylinders can be expanded (Remark 3.6). Hence $n = 4$ is the largest n for which the ring cannot be expanded.

1.4. Background. Kuperberg [10] posed Question 1.1 for $n = 6$ in 1990 (see also [4, Section 2.4, Conjecture 16]). In this case $\omega_6 = 1$, so the ball and the cylinders have the same radius. Heppes and Szabó [9] proved that $(8, 1) \in \mathcal{P}_{\text{cyl}}$. Braß and Wenk [5] improved this to $(7, 1) \in \mathcal{P}_{\text{cyl}}$. Yardimci [28] showed that seven cylinders of radius 1.04965 cannot touch the ball. Kuperberg's question was answered in [16, 15], which proves $(6, 1) \in \mathcal{P}_{\text{cyl}}$.

Kuperberg [10] also asked for the value of R_n [4, Section 2.4, Problem 17]. Ogievetsky and Shlosman conjectured that $R_4 = 1 + \sqrt{2}$ [18, 21, 17].

Firsching [7] found six cylinders of radius 1.049659 $> \omega_6$ around the unit ball and gave exact coordinates for them. His numerical search also found expansions of the rings of five and seven cylinders. Ogievetsky and Shlosman [18, 19, 20, 21] developed a general method for deforming the ring. They found a configuration C_m of six cylinders of radius $\frac{1}{8}(3 + \sqrt{33}) = 1.0930\dots$ and proved that it is a local maximum [19]. They showed that the ring can be expanded for every even $n \geq 6$ [18, Section 6]. The survey [22] describes this circle of ideas.

Our upper bounds use the two-point bound. It is the Lovász theta number [13] of a Cayley graph on the rotation group. De Laat and Vallentin [11] set up theta numbers for such infinite packing graphs. For packings of spherical caps the two-point bound is the linear programming bound of Delsarte, Goethals, and Seidel [6]. For packings on the rotation group it was first worked out by de Oliveira Filho, Spomer, and Vallentin [25]. They parametrize $\text{SO}(3)$ by Euler angles and use Bochner's theorem [1] to describe the functions of positive type. We use quaternions instead. Then every hypothesis of the bound becomes a statement about integers (Section 8). The surveys [23, 27] and the thesis [26] describe the history of these methods.

1.5. Methods. To prove $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$ we construct $n + 1$ cylinders of radius ω_n touching B . After the rescaling of Section 2, this means $n + 1$ lines tangent to the unit sphere with pairwise distances at least $2 \sin \frac{\pi}{n}$. For large n we use the *pinwheel* of Section 4. It consists of two groups of tangent lines at latitudes $\pm\varphi$, tilted by an angle δ and staggered in longitude. This is the family through which Ogievetsky and Shlosman unlock the ring [18, Section 6]. We fix one rational pair $(\sin \varphi, \tan \delta)$, so that every inequality becomes a comparison of rational numbers.

Theorem 1.8. *For every odd $n \geq 19$ and every even $n \geq 38$ we have $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$.*

The remaining 19 values $n \in \{9, 10, \dots, 18\} \cup \{20, 22, \dots, 36\}$ are treated in Section 6 by explicit configurations with rational data.

To prove $(n, \omega_n) \in \mathcal{P}_{\text{cyl}}$ we use the two-point bound on the rotation group $\text{SO}(3)$ [25, 27]. A finite list of rational data, called a *certificate*, proves each case $4 \leq n \leq 7$ of Theorem 1.5. The case $n = 3$ follows from Theorem 1.7, since

$\omega_3 > 1 + \sqrt{2}$. The certificates and the configurations are checked in exact arithmetic by a standalone verifier [14]. The proof of Theorem 1.7 is elementary and does not use a computer.

1.6. Organization. Section 2 replaces the cylinders by lines tangent to the unit sphere. Section 3 contains the elementary proof of the Ogievetsky–Shlosman conjecture (Theorem 1.7). Sections 4–6 construct $n + 1$ cylinders of radius ω_n for every $n \geq 9$. This proves that insertion is possible for these n (Theorem 1.6). Sections 7–12 prove that insertion is impossible for $3 \leq n \leq 7$ (Theorem 1.5). This part uses the two-point bound and rational certificates. Section 13 lists open problems, including the case $n = 8$. Appendix B describes the deposited data and the verifier.

2. THE CONFIGURATION SPACE AND THE SEPARATION PARAMETER

Fix $\omega > 0$ and let B be the unit ball centered at O . A cylinder of radius ω touches B if and only if its axis is tangent to the sphere of radius $1 + \omega$ centered at O . Two cylinders of radius ω are non-overlapping if and only if their axes are at distance at least 2ω . The homothety with center O and ratio $\frac{1}{1+\omega}$ maps the axes to lines tangent to the unit sphere $\mathbb{S}^2$ and multiplies all distances by $\frac{1}{1+\omega}$. This proves the following lemma, which we use throughout Sections 4–6.

Lemma 2.1. *Let $N \geq 2$ and $\omega > 0$. There exist N pairwise non-overlapping cylinders of radius ω touching B if and only if there exist N lines tangent to $\mathbb{S}^2$ with pairwise distances at least $\tau(\omega) = \frac{2\omega}{1+\omega}$.*

The map $\omega \mapsto \tau(\omega)$ is an increasing bijection of $(0, \infty)$ onto $(0, 2)$. So for $\omega > 0$ and $\tau \in (0, 2)$,

$$\tau = \frac{2\omega}{1+\omega} \iff \omega = \frac{\tau}{2-\tau}. \quad (3)$$

For $\tau \in (0, 2)$ we define

$$N_{\max}(\tau) = \mathcal{N}\left(\frac{\tau}{2-\tau}\right). \quad (4)$$

Remark 2.2. Since $\frac{\omega_n}{1+\omega_n} = \sin \frac{\pi}{n}$, the ring of n cylinders has separation $\tau(\omega_n) = 2 \sin \frac{\pi}{n}$. This is the edge length of a regular n -gon inscribed in the unit circle. Hence $N_{\max}(2 \sin \frac{\pi}{n}) \geq n$.

Let us recall the standard distance formula between lines (see, e.g., [18, Section 2]).

Lemma 2.3. *Let $\ell_i = \{\vec{p}_i + t\vec{v}_i : t \in \mathbb{R}\}$ for $i = 1, 2$. Then*

$$\text{dist}(\ell_1, \ell_2) = \begin{cases} |(\vec{p}_2 - \vec{p}_1) \cdot (\vec{v}_1 \times \vec{v}_2)| / \|\vec{v}_1 \times \vec{v}_2\|, & \text{if } \vec{v}_1 \nparallel \vec{v}_2, \\ \|(\vec{p}_2 - \vec{p}_1) \times \vec{v}_1\| / \|\vec{v}_1\|, & \text{if } \vec{v}_1 \parallel \vec{v}_2. \end{cases}$$

Let $\vec{e}_1, \vec{e}_2$, and $\vec{e}_3$ be the standard basis vectors of $\mathbb{R}^3$. Let $\ell_0 = \{\vec{e}_3 + t\vec{e}_1 : t \in \mathbb{R}\}$ be the line tangent to $\mathbb{S}^2$ at $\vec{e}_3$ with direction $\vec{e}_1$. The rotation group $\text{SO}(3)$ acts transitively on lines tangent to $\mathbb{S}^2$, so every such line is $g \cdot \ell_0$ for some $g \in \text{SO}(3)$, unique up to the stabilizer of ℓ_0 .

Definition 2.4. For $\tau \in (0, 2)$ we define the *safe zone* $\mathcal{D}(\tau)$ as the subset of $\text{SO}(3)$ given by

$$\mathcal{D}(\tau) = \{g \in \text{SO}(3) : \text{dist}(\ell_0, g \cdot \ell_0) \geq \tau\}.$$

When τ is fixed we write $\mathcal{D}$ for $\mathcal{D}(\tau)$.

Definition 2.5. A set $X \subseteq \text{SO}(3)$ will be called *non-overlapping* if $g^{-1}g' \in \mathcal{D}$ whenever g and g' are two distinct elements of X .

By (4) and Lemma 2.1, $N_{\max}(\tau)$ is the largest number of lines tangent to $\mathbb{S}^2$ with pairwise distances at least τ . Since rotations are isometries, $\{g_1, \dots, g_N\} \subset \text{SO}(3)$ is non-overlapping exactly when the lines $g_1 \cdot \ell_0, \dots, g_N \cdot \ell_0$ are distinct with pairwise distances at least τ . Hence $N_{\max}(\tau)$ is also the largest cardinality of a non-overlapping subset of $\text{SO}(3)$.

Lemma 2.6. *If $0 < \tau' \leq \tau < 2$, then $\mathcal{D}(\tau) \subseteq \mathcal{D}(\tau')$, and consequently $N_{\max}(\tau) \leq N_{\max}(\tau')$.*

Proof. The inclusion is immediate from Definition 2.4. Hence a set that is non-overlapping for τ is non-overlapping for τ' , so $N_{\max}(\tau) \leq N_{\max}(\tau')$. $\square$

Lemma 2.7. *The function $g \mapsto \text{dist}(\ell_0, g \cdot \ell_0)$ is upper semicontinuous on $\text{SO}(3)$.*

Proof. By definition $\text{dist}(\ell_0, g \cdot \ell_0) = \inf_{s,t \in \mathbb{R}} \|\vec{e}_3 + t\vec{e}_1 - g \cdot \vec{e}_3 - s g \cdot \vec{e}_1\|$, an infimum of a family of continuous functions of g , hence upper semicontinuous. $\square$

Corollary 2.8. *For every integer $N \geq 2$ the number $\tau^*(N) = \sup\{\tau \in (0, 2) : N_{\max}(\tau) \geq N\}$ is attained, in the sense that $N_{\max}(\tau) \geq N$ holds if and only if $\tau \leq \tau^*(N)$. Consequently $R_N = \frac{\tau^*(N)}{2 - \tau^*(N)}$ and $N(\omega) \geq N$ if and only if $\omega \leq R_N$.*

Proof. Let $\Phi(g_1, \dots, g_N) = \min_{i \neq j} \text{dist}(\ell_0, g_i^{-1} g_j \cdot \ell_0)$. Each of the finitely many entries is upper semicontinuous by Lemma 2.7, and a minimum of finitely many upper semicontinuous functions is upper semicontinuous. As $\text{SO}(3)^N$ is compact, Φ attains its maximum, say at $(\hat{g}_1, \dots, \hat{g}_N)$. Write $\hat{\tau} = \Phi(\hat{g}_1, \dots, \hat{g}_N)$.

If $\tau \leq \hat{\tau}$ then $\{\hat{g}_1, \dots, \hat{g}_N\}$ is non-overlapping for τ . Its elements are distinct because $\tau > 0$ and $\text{dist}(\ell_0, \ell_0) = 0$. So $N_{\max}(\tau) \geq N$. If $\tau > \hat{\tau}$ and $X = \{g_1, \dots, g_N\}$ were non-overlapping for τ , then $\Phi(g_1, \dots, g_N) \geq \tau > \hat{\tau}$, contradicting maximality. Hence $\tau^*(N) = \hat{\tau}$ and the stated equivalence holds. The last assertion follows by applying the increasing bijection (3). $\square$

3. PROOF OF THE OGIEVETSKY–SHLOSMAN CONJECTURE

Let ℓ be a line tangent to the unit sphere. Let $\vec{p}$ be the point of tangency, let $\vec{v}$ be a unit vector parallel to ℓ , and let $\vec{w} = \vec{p} \times \vec{v}$. We call $(\vec{p}, \vec{v}, \vec{w})$ an *orthonormal system associated to ℓ* . The only other system associated to ℓ is $(\vec{p}, -\vec{v}, -\vec{w})$. Passing from one to the other is called *reversing the direction of ℓ* .

Lemma 3.1. *Let ℓ and ℓ' be lines tangent to the unit sphere. Let $(\vec{p}, \vec{v}, \vec{w})$ and $(\vec{p}', \vec{v}', \vec{w}')$ be their associated orthonormal systems. Then*

$$2(\vec{p} \cdot \vec{p}')^2 - 2(\vec{w} \cdot \vec{w}')^2 = (\vec{v} \cdot \vec{w}' - \vec{w} \cdot \vec{v}')^2 + ((\vec{p}' - \vec{p}) \cdot (\vec{v} \times \vec{v}'))^2 - 2\|\vec{v} \times \vec{v}'\|^2. \quad (5)$$

Proof. Both sides of (5) are invariant under rotations. Hence we may assume that $\vec{p}', \vec{v}'$, and $\vec{w}'$ are the coordinate vectors $\vec{e}_1, \vec{e}_2$, and $\vec{e}_3$. Let $\vec{p} = \langle p_1, p_2, p_3 \rangle$, $\vec{v} = \langle v_1, v_2, v_3 \rangle$, and $\vec{w} = \langle w_1, w_2, w_3 \rangle$. Then $\vec{v} \times \vec{e}_2 = \langle -v_3, 0, v_1 \rangle$. The second coordinate of $\vec{p} \times \vec{v}$ is $w_2 = p_3 v_1 - p_1 v_3$. Therefore $(\vec{e}_1 - \vec{p}) \cdot (\vec{v} \times \vec{e}_2) = -(v_3 + w_2)$. The right-hand side of (5) becomes

$$(v_3 - w_2)^2 + (v_3 + w_2)^2 - 2(v_1^2 + v_3^2) = 2w_2^2 - 2v_1^2.$$

The coordinates of $\vec{e}_1$ in the orthonormal basis $\{\vec{p}, \vec{v}, \vec{w}\}$ are p_1, v_1 , and w_1 . Hence $p_1^2 + v_1^2 + w_1^2 = 1$. Since $\vec{w}$ is a unit vector, also $w_1^2 + w_2^2 + w_3^2 = 1$. Subtracting the two equations gives $w_2^2 - v_1^2 = p_1^2 - w_3^2$. This proves (5), because $\vec{p} \cdot \vec{p}' = p_1$ and $\vec{w} \cdot \vec{w}' = w_3$. $\square$

Lemma 3.2. *Let ℓ and ℓ' be lines tangent to the unit sphere. Let $(\vec{p}, \vec{v}, \vec{w})$ and $(\vec{p}', \vec{v}', \vec{w}')$ be their associated orthonormal systems. If $\text{dist}(\ell, \ell') \geq \sqrt{2}$, then $\vec{p} \cdot \vec{p}' \leq 0$ and $|\vec{w} \cdot \vec{w}'| \leq -\vec{p} \cdot \vec{p}'$. The last inequality is an equality if and only if the following two conditions hold.*

- (i) $\vec{v} \cdot \vec{w}' = \vec{w} \cdot \vec{v}'$.
- (ii) $\ell \parallel \ell'$ or $\text{dist}(\ell, \ell') = \sqrt{2}$.

Proof. The points $\vec{p}$ and $\vec{p}'$ lie on ℓ and ℓ' . Hence $\|\vec{p} - \vec{p}'\| \geq \text{dist}(\ell, \ell') \geq \sqrt{2}$. Since $\|\vec{p} - \vec{p}'\|^2 = 2 - 2\vec{p} \cdot \vec{p}'$, this means $\vec{p} \cdot \vec{p}' \leq 0$. If ℓ and ℓ' are not parallel, then Lemma 2.3 gives $((\vec{p}' - \vec{p}) \cdot (\vec{v} \times \vec{v}'))^2 = \text{dist}^2(\ell, \ell') \|\vec{v} \times \vec{v}'\|^2 \geq 2\|\vec{v} \times \vec{v}'\|^2$. If they are parallel, then $\vec{v} \times \vec{v}' = 0$. In both cases the last two terms on the right-hand side of (5) have a non-negative sum. The sum is zero if and only if (ii) holds. The first term is zero if and only if (i) holds. Lemma 3.1 now gives $(\vec{w} \cdot \vec{w}')^2 \leq (\vec{p} \cdot \vec{p}')^2$, with equality if and only if (i) and (ii) hold. It remains to recall that $\vec{p} \cdot \vec{p}' \leq 0$. $\square$

For $\beta \in [0, \pi)$ let $\vec{u} = \langle \cos \beta, \sin \beta, 0 \rangle$ and $\vec{u}' = \langle -\sin \beta, \cos \beta, 0 \rangle$. Let C_β be the configuration of the four tangent lines

$$\frac{1}{\sqrt{2}}(\pm \vec{e}_1 + \vec{e}_3) + \mathbb{R}\vec{e}_2 \quad \text{and} \quad \frac{1}{\sqrt{2}}(\pm \vec{u} - \vec{e}_3) + \mathbb{R}\vec{u}'. \quad (6)$$

It consists of two parallel pairs. The pairs lie on opposite sides of the plane $z = 0$ and their directions make the angle β .

Lemma 3.3. *For $\beta \in (0, \pi)$ all six pairwise distances in C_β are equal to $\sqrt{2}$. The configuration C_0 is the ring of four lines.*

Proof. The lines of the first pair are parallel. The difference of their base points is $\sqrt{2}\vec{e}_1$, which is orthogonal to $\vec{e}_2$. Hence their distance is $\sqrt{2}$. The same holds for the second pair. Now take one line from each pair. Then $\vec{e}_2 \times \vec{u}' = \sin \beta \vec{e}_3$. The difference of the base points has $\vec{e}_3$ -coordinate $-\sqrt{2}$. By Lemma 2.3 the distance is $\sqrt{2}$ when $\sin \beta \neq 0$. For $\beta = 0$ all four lines are parallel to $\vec{e}_2$, and their points of tangency form a square. $\square$

Theorem 3.4. *Let $\ell_1, \ell_2, \ell_3, \ell_4$ be lines tangent to the unit sphere. Their pairwise distances are at least $\sqrt{2}$ if and only if, after a rotation, they form C_β for some $\beta \in [0, \pi)$.*

Proof. The configurations C_β have the required distances by Lemma 3.3. Conversely, assume that the pairwise distances are at least $\sqrt{2}$. For each i let $(\vec{p}_i, \vec{v}_i, \vec{w}_i)$ be an associated orthonormal system. Let P and W be the Gram matrices with entries $P_{ij} = \vec{p}_i \cdot \vec{p}_j$ and $W_{ij} = \vec{w}_i \cdot \vec{w}_j$. Four vectors in $\mathbb{R}^3$ are linearly dependent. Hence there is a non-zero vector $\vec{a} \in \mathbb{R}^4$ with $\sum_i a_i \vec{w}_i = 0$. Reversing the direction of ℓ_i does not change the line, and it replaces $\vec{w}_i$ by $-\vec{w}_i$. We reverse every line with $a_i < 0$, and we replace each such a_i by $-a_i$. The relation $\sum_i a_i \vec{w}_i = 0$ still holds, and now $a_i \geq 0$ for all i . Lemma 3.2 gives

$$\begin{aligned} 0 &= \left\| \sum_i a_i \vec{w}_i \right\|^2 = \sum_i a_i^2 + \sum_{i \neq j} W_{ij} a_i a_j \geq \sum_i a_i^2 - \sum_{i \neq j} |W_{ij}| a_i a_j \\ &\geq \sum_i a_i^2 + \sum_{i \neq j} P_{ij} a_i a_j = \left\| \sum_i a_i \vec{p}_i \right\|^2 \geq 0. \end{aligned} \quad (7)$$

Hence all inequalities in (7) are equalities.

Let us prove the following claim:

The matrices W and P are equal, or become equal after reversing one more line. (8)

The equalities in (7) give $\sum_i a_i \vec{p}_i = 0$. They also give $W_{ij} = -|W_{ij}| = P_{ij}$ whenever $a_i > 0$ and $a_j > 0$. Now let $a_k = 0$. Then $\sum_i a_i P_{ik} = \vec{p}_k \cdot \sum_i a_i \vec{p}_i = \vec{p}_k \cdot 0 = 0$, and all terms $a_i P_{ik}$ are non-positive. Hence $P_{ik} = 0$ whenever $a_i > 0$. Lemma 3.2 gives $W_{ik} = 0 = P_{ik}$. The diagonal entries of W and P are all equal to 1. So far we know that $W_{ij} = P_{ij}$ for every pair $i \neq j$ in which at least one of a_i and a_j is positive. Since $\vec{a} \neq 0$ and $\vec{w}_i \neq 0$, at least two of the a_i are positive. Hence there is at most one pair (i, j) with $a_i = a_j = 0$. If there is none, the claim (8) is proved. Otherwise we may assume that $a_1, a_2 > 0$ and $a_3 = a_4 = 0$. Then $a_1 \vec{p}_1 = -a_2 \vec{p}_2$. Taking lengths gives $a_1 = a_2$, and hence $\vec{p}_2 = -\vec{p}_1$. In the same way $\sum_i a_i \vec{w}_i = 0$ gives $\vec{w}_2 = -\vec{w}_1$. Therefore $\vec{v}_2 = \vec{w}_2 \times \vec{p}_2 = \vec{w}_1 \times \vec{p}_1 = \vec{v}_1$. Let $k \in \{3, 4\}$. We showed above that $W_{1k} = P_{1k} = 0$ and $W_{2k} = P_{2k} = 0$. So the inequality of Lemma 3.2 is an equality for the pairs $(1, k)$ and $(2, k)$, and condition (i) gives

$$\vec{v}_1 \cdot \vec{w}_k = \vec{w}_1 \cdot \vec{v}_k = -\vec{w}_2 \cdot \vec{v}_k = -\vec{v}_2 \cdot \vec{w}_k = -\vec{v}_1 \cdot \vec{w}_k.$$

Hence $\vec{v}_1 \cdot \vec{w}_k = \vec{w}_1 \cdot \vec{v}_k = 0$. The vector $\vec{w}_1$ is orthogonal to $\vec{v}_k$ and, since $W_{1k} = 0$, also to $\vec{w}_k$. Therefore $\vec{p}_k = \pm \vec{w}_1$. The vector $\vec{w}_k$ is orthogonal to $\vec{v}_1$ and $\vec{w}_1$, so $\vec{w}_k = \pm \vec{p}_1$. Then $\vec{v}_k = \vec{w}_k \times \vec{p}_k = \pm \vec{p}_1 \times \vec{w}_1 = \pm \vec{v}_1$. If $\vec{p}_3 = \vec{p}_4$, then ℓ_3 and ℓ_4 have the same point of tangency and parallel directions, so they coincide. Hence $P_{34} = -1$ and $W_{34} = \pm 1$. Reversing ℓ_4 if necessary gives $W_{34} = P_{34}$. This proves the claim (8).

From now on $W = P$. The arrays of vectors $(\vec{p}_i)_{i=1}^4$ and $(\vec{w}_i)_{i=1}^4$ have the same Gram matrix. Hence the map $\sum_i x_i \vec{p}_i \mapsto \sum_i x_i \vec{w}_i$ is well defined and preserves lengths. It extends to an orthogonal matrix Q with $Q\vec{p}_i = \vec{w}_i$ for all i . If $\det Q = -1$, we reverse the directions of all four lines and replace Q by $-Q$. This does not change W . So Q is a rotation by an angle $\varphi \in [0, \pi]$ about a unit vector, which we may take to be $\vec{e}_3$. The matrix representation of Q is

$$Q = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

In particular $\sin \varphi \geq 0$. For every unit vector $\vec{x}$ it is straightforward to verify that the quadratic form associated with Q is

$$\vec{x} \cdot Q\vec{x} = \cos \varphi + (1 - \cos \varphi) (\vec{x} \cdot \vec{e}_3)^2.$$

Since $\vec{p}_i \cdot Q\vec{p}_i = \vec{p}_i \cdot \vec{w}_i = 0$, we get $\varphi \neq 0$, $\cos \varphi \leq 0$, and

$$\vec{p}_i = \langle r \cos \alpha_i, r \sin \alpha_i, \varepsilon_i h \rangle, \quad h = \sqrt{\frac{-\cos \varphi}{1 - \cos \varphi}}, \quad r = \sqrt{\frac{1}{1 - \cos \varphi}}, \quad \varepsilon_i \in \{1, -1\}.$$

Since $W = P$ and $P_{ij} \leq 0$, all pairs satisfy $|W_{ij}| = -P_{ij}$. By Lemma 3.2, $\vec{v}_i \cdot \vec{w}_j = \vec{w}_i \cdot \vec{v}_j$ for all $i \neq j$. Since $\vec{v}_i = \vec{w}_i \times \vec{p}_i$ and $\vec{v}_j = \vec{w}_j \times \vec{p}_j$, we have

$$\vec{v}_i \cdot \vec{w}_j - \vec{w}_i \cdot \vec{v}_j = \det(\vec{w}_i, \vec{p}_i, \vec{w}_j) - \det(\vec{w}_i, \vec{w}_j, \vec{p}_j) = -\det(\vec{w}_i, \vec{w}_j, \vec{p}_i + \vec{p}_j).$$

Hence the equation $\vec{v}_i \cdot \vec{w}_j = \vec{w}_i \cdot \vec{v}_j$ is equivalent to $\det(\vec{w}_i, \vec{w}_j, \vec{p}_i + \vec{p}_j) = 0$. Let $\delta = \alpha_i - \alpha_j$. Since $\vec{w}_i = Q\vec{p}_i = \langle r \cos(\alpha_i + \varphi), r \sin(\alpha_i + \varphi), \varepsilon_i h \rangle$, expanding the determinant gives

$$\begin{aligned} \det(\vec{w}_i, \vec{w}_j, \vec{p}_i + \vec{p}_j) &= \begin{vmatrix} r \cos(\alpha_i + \varphi) & r \cos(\alpha_j + \varphi) & r(\cos \alpha_i + \cos \alpha_j) \\ r \sin(\alpha_i + \varphi) & r \sin(\alpha_j + \varphi) & r(\sin \alpha_i + \sin \alpha_j) \\ \varepsilon_i h & \varepsilon_j h & (\varepsilon_i + \varepsilon_j)h \end{vmatrix} \\ &= r^2 h \cdot \begin{cases} -2\varepsilon_i(1 - \cos \varphi) \sin \delta, & \text{if } \varepsilon_i = \varepsilon_j, \\ -2\varepsilon_i \sin \varphi (1 + \cos \delta), & \text{if } \varepsilon_i \neq \varepsilon_j. \end{cases} \end{aligned}$$

Note that $r > 0$ and $1 - \cos \varphi > 0$. If $\cos \varphi = 0$, then $h = 0$. All $\vec{p}_i$ and $\vec{w}_i$ are orthogonal to $\vec{e}_3$, so all lines are parallel to $\vec{e}_3$. The four points $\vec{p}_i$ lie on a great circle, and their pairwise angles are at least $\frac{\pi}{2}$. Hence consecutive points are at angle exactly $\frac{\pi}{2}$. This is the ring C_0 .

If $-1 < \cos \varphi < 0$, then $h > 0$ and $\sin \varphi > 0$. Take two lines at the same height. The determinant vanishes, so $\sin \delta = 0$. Also $\delta \not\equiv 0 \pmod{2\pi}$, since otherwise the two lines coincide. So $\delta \equiv \pi \pmod{2\pi}$. Take two lines at different heights. The determinant vanishes, so $\cos \delta = -1$. Hence every two of the four angles α_i differ by π modulo 2π . This is impossible for three angles.

If $\cos \varphi = -1$, then $h = r = \frac{1}{\sqrt{2}}$ and $\sin \varphi = 0$. Two lines at the same height again satisfy $\delta \equiv \pi \pmod{2\pi}$. Hence each of the two circles contains exactly two of the points $\vec{p}_i$, and they are antipodal on that circle. After a rotation about $\vec{e}_3$, the points are $\frac{1}{\sqrt{2}}(\pm \vec{e}_1 + \vec{e}_3)$ and $\frac{1}{\sqrt{2}}(\pm \vec{u} - \vec{e}_3)$. Here Q is the rotation by π about $\vec{e}_3$. For $\vec{x} = \langle x_1, x_2, x_3 \rangle$ we get $Q\vec{x} \times \vec{x} = 2x_3 \langle -x_2, x_1, 0 \rangle$. So the directions are $\pm \vec{e}_2$ and $\pm \vec{u}'$. This is C_β . $\square$

Proof of Theorem 1.7. Suppose that four non-overlapping cylinders of radius $\omega' > 1 + \sqrt{2}$ touch the unit ball. By Lemma 2.1 there are four tangent lines with pairwise distances at least $\tau(\omega') > \sqrt{2}$. By Theorem 3.4 they form a configuration C_β . By Lemma 3.3 two of them are at distance exactly $\sqrt{2}$. This is a contradiction. $\square$

Remark 3.5. The homothety of Lemma 2.1 maps configurations of cylinders bijectively to configurations of tangent lines. Hence Theorem 3.4 proves the conjecture of Ogievetsky and Shlosman [18, 21] and [17, Section 3.1] in full. They conjectured that the four-cylinder configurations of radius $1 + \sqrt{2}$ form a circle modulo rotations, and that no configuration of larger radius exists. The family C_β is the motion found in [18, Section 6.3]. One parallel pair stays fixed and the other pair rotates about the common perpendicular $\vec{e}_3$. As $\beta \rightarrow \pi$, the configuration C_β tends to C_0 . Measure the angle from the direction of the first pair to the direction of the second pair about the axis that joins the pairs. This angle is β , and it does not change under rotations. Hence distinct $\beta \in [0, \pi)$ give configurations that are not rotations of each other, and the family is a circle.

Remark 3.6. The ring of five cylinders can be expanded. A numerical configuration with this property was found by Firsching [7]. The standalone verifier [14] confirms it in exact arithmetic for a rational configuration of five tangent lines (Appendix B).

4. THE PINWHEEL AND ITS DISTANCE FORMULAS

Definition 4.1 (Pinwheel). Let $N = 2k \geq 6$ be even and set $\gamma = \frac{2\pi}{k} = \frac{4\pi}{N}$. For a latitude $\mu \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and longitude λ , let

$$x(\mu, \lambda) = (\cos \mu \cos \lambda, \cos \mu \sin \lambda, \sin \mu) \in \mathbb{S}^2,$$

and let $\vec{u} = \langle -\sin \mu \cos \lambda, -\sin \mu \sin \lambda, \cos \mu \rangle$ and $\vec{t} = \langle -\sin \lambda, \cos \lambda, 0 \rangle$ be the unit tangent vectors at $x(\mu, \lambda)$ pointing north and east. Given parameters $\varphi \in (0, \frac{\pi}{2})$, $\delta \in (0, \frac{\pi}{2})$ and $\psi \in \mathbb{R}$, the pinwheel $P(N; \varphi, \delta, \psi)$ is the family

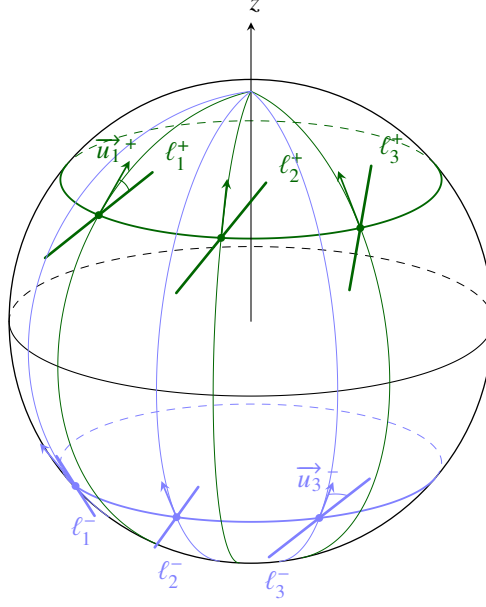


FIGURE 2. A few of the lines of a pinwheel configuration. Three tangent points lie on the circle of latitude φ (dark green) and three on the circle of latitude $-\varphi$ (light blue). Each meridian is drawn in the colour of the group of its tangent point. The two groups are staggered in longitude by the free offset ψ . At each tangent point a short piece of the corresponding tangent line is drawn. The arrow at each tangent point is the north-pointing unit tangent vector, i.e. the direction of increasing latitude. Note that it points towards the north pole on the southern circle as well. Every line is obtained by rotating this vector eastward by the same angle δ within the tangent plane. The small arcs at ℓ_1^+ and ℓ_3^- mark this angle.

of N lines tangent to $\mathbb{S}^2$ consisting of the *up group*

$$\ell_j^+ : \text{ tangent point } x(\varphi, j\gamma), \quad \text{direction } \vec{v} = \cos \delta \vec{u} + \sin \delta \vec{t}, \quad j = 0, \dots, k-1,$$

and the *down group*

$$\ell_j^- : \text{ tangent point } x(-\varphi, j\gamma + \psi), \quad \text{direction } \vec{v} = \cos \delta \vec{u} + \sin \delta \vec{t}, \quad j = 0, \dots, k-1.$$

Up to a rotation about the vertical axis and a reflection, $P(N; \varphi, \delta, \psi)$ is the configuration $C_{N/2}(\varphi, \kappa, \delta)$ of Ogievetsky and Shlosman [18, Section 6] with $\psi = \frac{\gamma}{2} + 2\kappa$. We index by the number of lines rather than by the number of lines in one group.

4.1. The two distance formulas. By the rotational symmetry about the z -axis it suffices to compute the distance between a tangent line at latitude φ and longitude 0 and a tangent line at latitude $\pm\varphi$ and longitude Δ .

Let $\varphi \in (0, \frac{\pi}{2})$ and $\Delta \in [0, 2\pi)$. Let P_1 be the point of the unit sphere with latitude φ and longitude 0, and $P_2^\pm$ the point with latitude $\pm\varphi$ and longitude Δ , so that

$$\begin{aligned} \vec{p}_1 &= \langle \cos \varphi, 0, \sin \varphi \rangle, \\ \vec{p}_2^\pm &= \langle \cos \Delta \cos \varphi, \sin \Delta \cos \varphi, \pm \sin \varphi \rangle. \end{aligned}$$

Let $\vec{u}_1$ and $\vec{u}_2^+$ be the unit tangent vectors at P_1 and P_2^+ pointing north, namely

$$\begin{aligned} \vec{u}_1 &= \langle -\sin \varphi, 0, \cos \varphi \rangle; \\ \vec{u}_2^+ &= \langle -\cos \Delta \sin \varphi, -\sin \Delta \sin \varphi, \cos \varphi \rangle. \end{aligned}$$

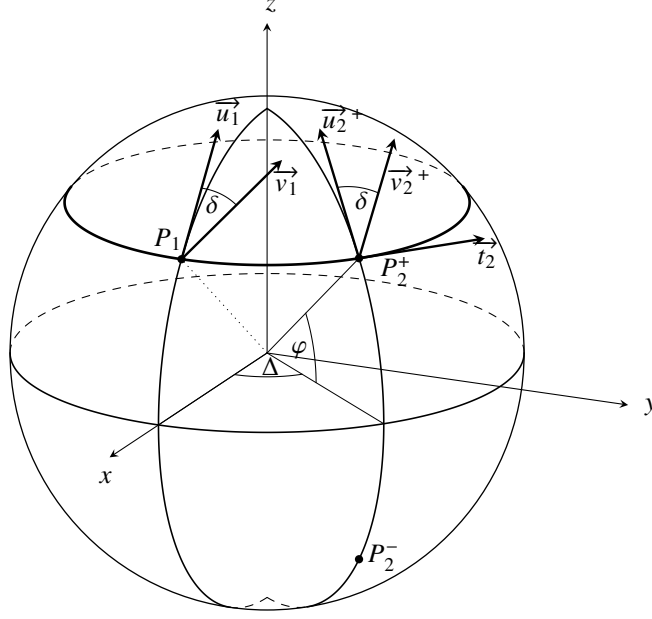


FIGURE 3. The points $P_1, P_2^\pm$ and the tangent frames from Lemma 4.2. The thick circle is the latitude circle at latitude φ containing P_1 and P_2^+ . The meridians through P_1 and through $P_2^\pm$ and the equator are drawn with hidden parts dashed. The vector $\vec{u}_1$ lies along the meridian through P_1 , as $\vec{u}_2^+$ does along the meridian through P_2^+ . The vectors $\vec{u}_1$ and $\vec{u}_2^+$ point in the direction of increasing latitude, $\vec{t}_2$ points eastward, and $\vec{v}_1, \vec{v}_2^+$ are obtained by rotating $\vec{u}_1, \vec{u}_2^+$ by the angle δ within the respective tangent planes, as in (9). The vector $\vec{v}_2^-$ at P_2^- (not drawn) is obtained from (11).

Let $\vec{t}_2 = \langle -\sin \Delta, \cos \Delta, 0 \rangle$ be the unit tangent vector at P_2^+ pointing east. Then $\{\vec{t}_2, \vec{u}_2^+\}$ is an orthonormal basis of the tangent plane at P_2^+ . For $\delta \in (0, \frac{\pi}{2})$ let

$$\begin{aligned} \vec{v}_2^+ &= \cos \delta \vec{u}_2^+ + \sin \delta \vec{t}_2 \\ &= \langle -\sin \delta \sin \Delta - \cos \delta \cos \Delta \sin \varphi, \sin \delta \cos \Delta - \cos \delta \sin \Delta \sin \varphi, \cos \delta \cos \varphi \rangle. \end{aligned} \quad (9)$$

Setting $\Delta = 0$ in (9) gives the corresponding vector at P_1 ,

$$\vec{v}_1 = \langle -\sin \varphi \cos \delta, \sin \delta, \cos \varphi \cos \delta \rangle, \quad (10)$$

and replacing φ by $-\varphi$ in (9) gives the corresponding vector at P_2^- ,

$$\vec{v}_2^- = \langle -\sin \delta \sin \Delta + \sin \varphi \cos \delta \cos \Delta, \sin \delta \cos \Delta + \sin \varphi \cos \delta \sin \Delta, \cos \varphi \cos \delta \rangle. \quad (11)$$

Let $\ell^+(0)$, $\ell^+(\Delta)$ and $\ell^-(\Delta)$ be the lines through P_1 , P_2^+ and P_2^- with directions $\vec{v}_1$, $\vec{v}_2^+$ and $\vec{v}_2^-$. The argument records the longitude. The lines of Definition 4.1 are $\ell_j^+ = \ell^+(j\gamma)$ and $\ell_j^- = \ell^-(j\gamma + \psi)$.

Lemma 4.2. *Assume that $\varphi \in (0, \frac{\pi}{2})$, $\Delta \in [0, 2\pi)$, and $\delta \in (0, \frac{\pi}{2})$. The squares of the distances between $\ell^+(0)$ and $\ell^\pm(\Delta)$ satisfy*

$$\text{dist}^2(\ell^+(0), \ell^+(\Delta)) = \frac{4 \cos^4 \varphi \tan^2 \delta \sin^2 \frac{\Delta}{2}}{\left(\tan^2 \delta + \sin^2 \varphi \right) \cdot \left[1 + \tan^2 \delta - \sin^2 \frac{\Delta}{2} \left(\tan^2 \delta + \sin^2 \varphi \right) \right]}; \quad (12)$$

$$\text{dist}^2(\ell^+(0), \ell^-(\Delta)) = \frac{4 \left(\sin \varphi \tan \delta + \tan \frac{\Delta}{2} \right)^2}{\cos^2 \varphi (1 + \tan^2 \delta) + \left(\sin \varphi \tan \delta + \tan \frac{\Delta}{2} \right)^2} \quad \text{for } \Delta \neq \pi, \quad (13)$$

while $\text{dist}(\ell^+(0), \ell^-(\Delta)) = 2$ for $\Delta = \pi$.

Proof. If $\vec{v}_1$ and $\vec{v}_2^\pm$ are not parallel, then

$$\text{dist}(\ell^+(0), \ell^\pm(\Delta)) = \frac{|(\vec{p}_2^\pm - \vec{p}_1) \cdot (\vec{v}_1 \times \vec{v}_2^\pm)|}{\|\vec{v}_1 \times \vec{v}_2^\pm\|}. \quad (14)$$

Write $\vec{v}_1 \times \vec{v}_2^\pm = \langle N_x^\pm, N_y^\pm, N_z^\pm \rangle$. From (9)–(11),

$$\begin{aligned} N_x^\pm &= (\sin \delta (1 - \cos \Delta) \pm \cos \delta \sin \varphi \sin \Delta) \cos \delta \cos \varphi, \\ N_y^\pm &= (\cos \delta \sin \varphi (1 \mp \cos \Delta) - \sin \delta \sin \Delta) \cos \delta \cos \varphi, \\ N_z^\pm &= (-1 \pm 1) \sin \delta \cos \delta \sin \varphi \cos \Delta + \sin \Delta (\sin^2 \delta \pm \cos^2 \delta \sin^2 \varphi). \end{aligned}$$

The line $\ell^+(\Delta)$. Here $\vec{p}_2^+ - \vec{p}_1 = \cos \varphi \langle \cos \Delta - 1, \sin \Delta, 0 \rangle$, and expanding,

$$\begin{aligned} (\vec{p}_2^+ - \vec{p}_1) \cdot (\vec{v}_1 \times \vec{v}_2^+) &= -2 \cos^2 \varphi \sin \delta \cos \delta (1 - \cos \Delta), \\ \|\vec{v}_1 \times \vec{v}_2^+\|^2 &= (\sin^2 \delta + \cos^2 \delta \sin^2 \varphi) \\ &\quad \cdot \left(2(1 - \cos \Delta) \cos^2 \delta \cos^2 \varphi + \sin^2 \Delta (\sin^2 \delta + \cos^2 \delta \sin^2 \varphi) \right). \end{aligned}$$

For $\Delta \in (0, 2\pi)$ the norm is positive, since its second factor is at least $2(1 - \cos \Delta) \cos^2 \delta \cos^2 \varphi > 0$, so (14) applies. For $\Delta = 0$ the lines coincide and both sides of (12) vanish. Substituting, dividing numerator and denominator by $\cos^2 \delta$, and using $1 - \cos^2 \Delta = (1 - \cos \Delta)(1 + \cos \Delta)$, $1 - \cos \Delta = 2 \sin^2 \frac{\Delta}{2}$ and $\cos^2 \varphi + \tan^2 \delta + \sin^2 \varphi = 1 + \tan^2 \delta$ gives (12). The bracket in its denominator is at least $1 + \tan^2 \delta - (\tan^2 \delta + \sin^2 \varphi) = \cos^2 \varphi > 0$.

The line $\ell^-(\Delta)$. Here $\vec{p}_2^- - \vec{p}_1 = \langle \cos \varphi (\cos \Delta - 1), \cos \varphi \sin \Delta, -2 \sin \varphi \rangle$. With the abbreviations

$$\begin{aligned} A &= \sin \varphi \cos \delta \cos \frac{\Delta}{2} - \sin \delta \sin \frac{\Delta}{2}, \\ B &= \sin \delta \sin \varphi \cos \frac{\Delta}{2} + \cos \delta \sin \frac{\Delta}{2}, \\ C &= \sin \varphi \cos \delta \sin \frac{\Delta}{2} + \sin \delta \cos \frac{\Delta}{2}, \end{aligned} \quad (15)$$

a direct expansion, using the half-angle formulas, gives

$$(\vec{p}_2^- - \vec{p}_1) \cdot (\vec{v}_1 \times \vec{v}_2^-) = 4AB, \quad N_z^- = -2AC, \quad (N_x^-)^2 + (N_y^-)^2 = 4 \cos^2 \delta \cos^2 \varphi A^2, \quad (16)$$

so $\|\vec{v}_1 \times \vec{v}_2^-\|^2 = 4A^2 [\cos^2 \delta \cos^2 \varphi + C^2]$. The mixed terms in $A^2 + C^2$ cancel, whence

$$A^2 + C^2 = \sin^2 \varphi \cos^2 \delta + \sin^2 \delta, \quad \text{i.e.} \quad \cos^2 \delta \cos^2 \varphi + A^2 + C^2 = 1. \quad (17)$$

In particular $\cos^2 \delta \cos^2 \varphi + C^2 \geq \cos^2 \delta \cos^2 \varphi > 0$, and the two direction vectors are parallel if and only if $A = 0$.

If $A \neq 0$, then (14) and (16) give

$$\text{dist}^2(\ell^+(0), \ell^-(\Delta)) = \frac{4B^2}{\cos^2 \delta \cos^2 \varphi + C^2}. \quad (18)$$

For $\Delta = \pi$ (where $A = -\sin \delta \neq 0$) we have $B = \cos \delta$, $C = \sin \varphi \cos \delta$, and (18) equals 4, proving the last claim. For $\Delta \neq \pi$, dividing numerator and denominator of (18) by $\cos^2 \delta \cos^2 \frac{\Delta}{2} > 0$ and using $\tan^2 \delta = \cos^2 \varphi \tan^2 \delta + \sin^2 \varphi \tan^2 \delta$ turns the denominator into $\cos^2 \varphi (1 + \tan^2 \delta) + (\sin \varphi \tan \delta + \tan \frac{\Delta}{2})^2$, which is (13).

If $A = 0$, the parallel unit vectors $\vec{v}_1, \vec{v}_2^-$ have the same positive third coordinate $\cos \delta \cos \varphi$, so they are equal. Moreover $(\vec{p}_2^- - \vec{p}_1) \cdot \vec{v}_1 = -2 \cos \varphi \cos \frac{\Delta}{2} A = 0$, so the distance between the two parallel lines is $\|\vec{p}_2^- - \vec{p}_1\|$, whose square $2 \cos^2 \varphi (1 - \cos \Delta) + 4 \sin^2 \varphi = 4(A^2 + B^2) = 4B^2$ agrees with (18), since (17) with $A = 0$ gives $\cos^2 \delta \cos^2 \varphi + C^2 = 1$. Finally $A = 0$ forces $\tan \frac{\Delta}{2} = \sin \varphi \cot \delta > 0$, so $\Delta \neq \pi$ and the reduction to (13) applies verbatim. $\square$

4.2. The formulas in S and T . Throughout the pinwheel construction we abbreviate

$$S = \sin \varphi \in (0, 1), \quad T = \tan \delta > 0, \quad c_0 = (1 - S^2)(1 + T^2).$$

By Lemma 4.2, every distance in the pinwheel is an algebraic function of S and T . In Section 5 we choose S and T rational, so that the inequalities we need become comparisons of rational numbers.

For $\Delta \in \mathbb{R}$ let $D_{\text{same}}(\Delta) = \text{dist}(\ell^+(0), \ell^+(\Delta))$ and $D_{\text{cross}}(\Delta) = \text{dist}(\ell^+(0), \ell^-(\Delta))$. With $\cos^4 \varphi = (1 - S^2)^2$, (12) reads

$$D_{\text{same}}(\Delta)^2 = \frac{4 \sin^2 \frac{\Delta}{2} (1 - S^2)^2 T^2}{(S^2 + T^2)(1 + T^2 - \sin^2 \frac{\Delta}{2} (S^2 + T^2))}, \quad (19)$$

and the bracket in the denominator is at least $1 - S^2 > 0$, as shown in the proof of Lemma 4.2. For $\Delta \neq \pi$ write $w = \tan \frac{\Delta}{2} + ST$. The identity $\cos^2 \varphi (1 + \tan^2 \delta) = c_0$ turns (13) into

$$D_{\text{cross}}(\Delta) = \frac{2|w|}{\sqrt{w^2 + c_0}}, \quad \Delta \neq \pi. \quad (20)$$

By Lemma 4.2, $D_{\text{cross}}(\pi) = 2$. This is also the limit of (20) as $|w| \rightarrow \infty$. Both formulas are 2π -periodic in Δ . For (20) we take $\Delta \in (-\pi, \pi]$, on which $\Delta \mapsto \tan \frac{\Delta}{2}$ is an increasing bijection of $(-\pi, \pi)$ onto $\mathbb{R}$.

The two formulas cover all pairs of pinwheel lines. A rotation about the z -axis carries two up lines with longitudes λ_1, λ_2 to the pair $(\ell^+(0), \ell^+(\Delta))$ with $\Delta = \lambda_2 - \lambda_1$, and carries an up line and a down line to the pair $(\ell^+(0), \ell^-(\Delta))$ with Δ equal to the longitude of the down line minus that of the up line. The reflection $(x, y, z) \mapsto (x, y, -z)$ carries two down lines to two lines of up type with the same longitudes and tilt $-\delta$, and (19) depends on δ only through T^2 . Hence two lines of the same group with longitudes λ_1, λ_2 are at distance $D_{\text{same}}(\lambda_2 - \lambda_1)$, and an up line and a down line whose longitudes differ by Δ are at distance $D_{\text{cross}}(\Delta)$.

Lemma 4.3. *As a function of $x = \sin^2 \frac{\Delta}{2} \in (0, 1]$, the right-hand side of (19) is strictly increasing. Consequently, within either group of $P(N; \varphi, \delta, \psi)$ the minimum distance is attained by adjacent lines and equals $D_{\text{same}}(\gamma)$.*

Proof. The right-hand side is $\frac{Ax}{B-Cx}$ with $A = 4(1 - S^2)^2 T^2 > 0$, $B = (S^2 + T^2)(1 + T^2)$, $C = (S^2 + T^2)^2$, and $B - Cx \geq (S^2 + T^2)(1 - S^2) > 0$ on $[0, 1]$ by the bound recorded after (19). Such a function is strictly increasing. Within a group the longitude differences are $\Delta = j\gamma$, $1 \leq j \leq k - 1$, and $\frac{j\gamma}{2} \in [\frac{\gamma}{2}, \pi - \frac{\gamma}{2}]$, so $\sin^2 \frac{j\gamma}{2} \geq \sin^2 \frac{\gamma}{2}$, with equality for $j \in \{1, k - 1\}$. $\square$

4.3. Criterion in S and T . By Definition 4.1, the longitude of ℓ_i^- minus the longitude of ℓ_j^+ is

$$\Delta_{ij} = \psi + (i - j)\gamma \pmod{2\pi},$$

so the cross-group longitude differences form the set

$$\{\psi + j\gamma \pmod{2\pi} : j = 0, 1, \dots, k - 1\}, \quad (21)$$

an arithmetic progression of step γ on the circle whose offset ψ we are free to choose.

Lemma 4.4. *Let $\tau^* = 2 \sin \theta$ with $\theta \in (0, \frac{\pi}{2})$, and set*

$$u^* = \sqrt{c_0} \frac{\tau^*}{\sqrt{4 - \tau^{*2}}} = \sqrt{c_0} \tan \theta. \quad (22)$$

Let

$$J = \left(2 \arctan(-ST - u^*), 2 \arctan(-ST + u^*) \right).$$

Then for $\Delta \in (-\pi, \pi]$ we have $D_{\text{cross}}(\Delta) < \tau^$ if and only if $\Delta \in J$. The open interval J has length*

$$|J| = 2 \left[\arctan(ST + u^*) - \arctan(ST - u^*) \right].$$

Let

$$\psi^* = \frac{\gamma}{2} - \arctan(ST - u^*) - \arctan(ST + u^*). \quad (23)$$

If $|J| \leq \gamma$ and $\psi = \psi^$, then the lattice (21) is disjoint from J . Hence every cross-group distance in $P(N; \varphi, \delta, \psi^*)$ is at least τ^* .*

Proof. The function $t \mapsto \frac{2t}{\sqrt{t^2 + c_0}}$ is strictly increasing on $[0, \infty)$, and u^* of (22) is the value at which it equals τ^* .

Hence $D_{\text{cross}}(\Delta) < \tau^*$ if and only if $|w| < u^*$. Now $|w| < u^*$ means $\tan \frac{\Delta}{2} \in (-ST - u^*, -ST + u^*)$, and as $\Delta \mapsto \tan \frac{\Delta}{2}$ is an increasing bijection from $(-\pi, \pi)$ onto $\mathbb{R}$ (and $D_{\text{cross}}(\pi) = 2 \geq \tau^*$), the failure set is exactly the interval J . By the oddness of $\arctan$,

$$|J| = 2 \left[\arctan(u^* - ST) + \arctan(u^* + ST) \right] = 2 \left[\arctan(ST + u^*) - \arctan(ST - u^*) \right].$$

The interval J is centered at $c = \arctan(-ST - u^*) + \arctan(-ST + u^*) = \psi^* - \frac{\gamma}{2}$. With $\psi = \psi^*$ the lattice (21) is $\{c + \frac{\gamma}{2} + j\gamma\}$, whose distance (on the circle) from the center c is at least $\frac{\gamma}{2} \geq \frac{|J|}{2}$. Since J is open, no lattice point lies in J . $\square$

Fix an even $N \geq 6$, parameters $S \in (0, 1)$, $T > 0$, and a target $\tau^* = 2 \sin \theta$ with $\theta \in (0, \frac{\pi}{2})$. By Lemma 4.3, all same-group distances are at least τ^* if and only if $D_{\text{same}}(\frac{4\pi}{N})^2 \geq \tau^{*2}$. Multiplied out, this *same-group inequality* reads

$$\sin^2 \frac{2\pi}{N} (1 - S^2)^2 T^2 \geq \sin^2 \theta \left[(S^2 + T^2)(1 + T^2) - \sin^2 \frac{2\pi}{N} (S^2 + T^2)^2 \right]. \quad (\text{same-group})$$

Lemma 4.4 shows that all cross-group distances are at least τ^* once the offset is chosen by (23), provided the interval J on which $D_{\text{cross}} < \tau^*$ is no longer than $\frac{4\pi}{N}$. This is the *cross-group inequality*

$$\arctan(ST + \sqrt{c_0} \tan \theta) - \arctan(ST - \sqrt{c_0} \tan \theta) \leq \frac{2\pi}{N}. \quad (\text{cross-group})$$

Theorem 4.5. *Let $N = 2k \geq 6$ be even, $S \in (0, 1)$, $T > 0$, and $\tau^* = 2 \sin \theta$ with $\theta \in (0, \frac{\pi}{2})$. If (same-group) and (cross-group) hold, then the pinwheel $P(N; \varphi, \delta, \psi^*)$ with $\sin \varphi = S$, $\tan \delta = T$ and ψ^* as in (23) consists of N lines tangent to $\mathbb{S}^2$ with all pairwise distances at least τ^* . Consequently, by Lemma 2.1, at least N pairwise non-overlapping cylinders of radius $\frac{\sin \theta}{1 - \sin \theta}$ touch the unit ball.*

Proof. Same-group pairs: by Lemma 4.3 the minimum is $D_{\text{same}}(\gamma)$ with $\gamma = \frac{4\pi}{N}$, and (same-group) is exactly $D_{\text{same}}(\gamma)^2 \geq \tau^{*2}$ after multiplying out the (positive) denominator of (19) at $\sin^2 \frac{\gamma}{2} = \sin^2 \frac{2\pi}{N}$. Cross-group pairs: (cross-group) is $|J| \leq \gamma$, so Lemma 4.4 applies. The cylinder statement follows since $\tau^* = 2 \sin \theta$ corresponds under (3) to $\omega = \frac{\sin \theta}{1 - \sin \theta}$. $\square$

Remark 4.6. Taking $\theta = \frac{\pi}{N-1}$, the cylinders in Theorem 4.5 have radius exactly ω_{N-1} , and $N = (N-1) + 1$ of them touch B . Taking $\theta = \frac{\pi}{N-2}$ gives $N = (N-2) + 2$ cylinders of radius exactly ω_{N-2} . These are the two cases used in Section 5.

5. THE PINWHEEL FOR LARGE n : PROOF OF THEOREM 1.8

5.1. The frozen parameters. From now on $S = \frac{9}{20}$, $T = \frac{31}{64}$, so that all derived constants are rational:

$$ST = \frac{279}{1280}, \quad S^2 T^2 = \frac{77841}{1638400}, \quad c_0 = (1 - S^2)(1 + T^2) = \frac{1613183}{1638400}.$$

We will need

$$2\Phi > \frac{21}{20} \quad \text{and} \quad \frac{21}{20} \sqrt{c_0} < 1 + S^2 T^2, \quad (24)$$

$$\text{where } \Phi \text{ is defined by } \Phi^2 = \frac{(1 - S^2)^2 T^2}{(S^2 + T^2)(1 + T^2)} = \frac{1564677136}{5658909425}.$$

After squaring, both inequalities in (24) are comparisons of rational numbers, checked by the verifier (Table 1).

5.2. Elementary estimates. Appendix A collects, with one-line proofs, the only analytic inequalities we use:

- (E1) $x - \frac{x^3}{6} \leq \sin x \leq x$ for $x \geq 0$;
- (E2) $\tan y \leq y + y^3$ for $y \in (0, 1]$;
- (E3) $\arctan p - \arctan q = \arctan \frac{p-q}{1+pq}$ whenever $pq > -1$, and $0 \leq \arctan z \leq z$ for $z \geq 0$;
- (E4) $z \mapsto \frac{z}{1 + S^2 T^2 - z^2}$ is strictly increasing on $[0, \sqrt{1 + S^2 T^2})$.

5.3. The two tail lemmas. Fix $s \in \{1, 2\}$ and put $\theta = \theta(N) = \frac{\pi}{N-s}$ in (same-group)–(cross-group).

Lemma 5.1. *Let $N \geq 2s + 4$ be even. If*

$$2\left(1 - \frac{s}{N}\right)\left(1 - \frac{2\pi^2}{3N^2}\right) \geq \frac{1}{\Phi}, \quad (25)$$

then (same-group) holds with $\theta = \frac{\pi}{N-s}$. The left-hand side of (25) is strictly increasing in N , and (25) holds for $s = 1$, $N = 32$ and for $s = 2$, $N = 48$. Hence (same-group) holds for all even $N \geq 32$ (respectively $N \geq 48$).

inequality	checked at	used in
(same-group), (cross-group), $\theta = \frac{\pi}{N-1}$	even N , $20 \leq N \leq 30$	Theorem 1.8
(same-group), (cross-group), $\theta = \frac{\pi}{N-2}$	even N , $40 \leq N \leq 46$	Theorem 1.8
(26)	$(s, N) = (1, 32)$ and $(2, 48)$	Lemma 5.2
(24)	independent of N	Lemma 5.1

TABLE 1. The numerical inequalities of the pinwheel construction, all checked by the verifier.

Proof. Discarding the subtracted term on the right of (same-group) shows that $\sin \frac{2\pi}{N} \Phi \geq \sin \theta$ is sufficient. By (E1), $\sin \frac{2\pi}{N} \geq \frac{2\pi}{N} (1 - \frac{2\pi^2}{3N^2})$ and $\sin \theta \leq \theta = \frac{\pi}{N-s}$, so (25) suffices. The left side of (25) is a product of two positive factors, each strictly increasing in N . For the two threshold cases, (24) gives $\frac{1}{\Phi} < \frac{40}{21} < 1.905$. Using $\pi^2 < 10$, the left-hand side of (25) is at least $\frac{31}{16} \cdot \frac{763}{768} > 1.92$ at $(s, N) = (1, 32)$ and at least $\frac{23}{12} \cdot \frac{1723}{1728} > 1.91$ at $(s, N) = (2, 48)$. $\square$

Lemma 5.2. Let $N \geq 2s + 4$ be even, $\theta = \frac{\pi}{N-s} \leq 1$, and set $\bar{u} = \sqrt{c_0}(\theta + \theta^3)$. If $\bar{u}^2 < 1 + S^2 T^2$ and

$$N \bar{u} \leq \pi(1 + S^2 T^2 - \bar{u}^2), \quad (26)$$

then (cross-group) holds with $\theta = \frac{\pi}{N-s}$. The left-hand side of (26) is strictly decreasing in N and the right-hand side is increasing, so (26), once true, remains true for all larger N . It holds for $s = 1$, $N = 32$ and for $s = 2$, $N = 48$, hence for all even $N \geq 32$ (respectively $N \geq 48$).

Proof. By (E2), $u^* = \sqrt{c_0} \tan \theta \leq \bar{u}$, so $u^{*2} \leq \bar{u}^2 < 1 + S^2 T^2$. By (E3) (the product $(ST + u^*)(ST - u^*) = S^2 T^2 - u^{*2} > -1$),

$$\begin{aligned} \arctan(ST + u^*) - \arctan(ST - u^*) &= \arctan \frac{2u^*}{1 + S^2 T^2 - u^{*2}} \leq \frac{2u^*}{1 + S^2 T^2 - u^{*2}} \\ &\leq \frac{2\bar{u}}{1 + S^2 T^2 - \bar{u}^2} \leq \frac{2\pi}{N}, \end{aligned}$$

using (E4) and (26). For the monotonicity, $N\bar{u} = \pi\sqrt{c_0} \frac{N}{N-s} (1 + \theta^2)$ is a product of positive strictly decreasing factors of N , while $\bar{u}$ itself decreases, so $\pi(1 + S^2 T^2 - \bar{u}^2)$ increases. The two threshold cases are in Table 1. $\square$

5.4. The computer-checked inequalities. Table 1 lists every numerical inequality that the pinwheel construction relies on. In each of them, the ingredients π , $\sin$, $\tan$, $\arctan$ and $\sqrt{c_0}$ are replaced by rational bounds. Each bound is chosen on the side that makes the resulting inequality imply the original one. What remains is a comparison of two explicit rational numbers. The standalone verifier [14], invoked as `cylie -i constructions`, performs all of them in exact arithmetic, with 512-bit overflow-checked integers and no floating point. The same run performs the checks (W1)–(W3) of Section 6 and ends with the line `ALL CONSTRUCTIONS VERIFIED`.

5.5. Proof of Theorem 1.8. Let $n \geq 19$ be odd and $N = n + 1$, so N is even, $N \geq 20$, and $\theta = \frac{\pi}{n} = \frac{\pi}{N-1}$. If $20 \leq N \leq 30$, inequalities (same-group) and (cross-group) hold by Table 1. If $N \geq 32$, the inequalities hold by Lemmas 5.1 and 5.2. By Theorem 4.5 and Remark 4.6, $N = n + 1$ pairwise non-overlapping cylinders of radius ω_n touch B , so $N(\omega_n) \geq n + 1$.

Let $n \geq 38$ be even and $N = n + 2$, so N is even, $N \geq 40$, and $\theta = \frac{\pi}{n} = \frac{\pi}{N-2}$. If $40 \leq N \leq 46$, use Table 1. If $N \geq 48$, use Lemmas 5.1 and 5.2. As before, $N(\omega_n) \geq n + 2$. $\square$

6. THE WINDOW: EXPLICIT CONFIGURATIONS FOR $9 \leq n \leq 36$

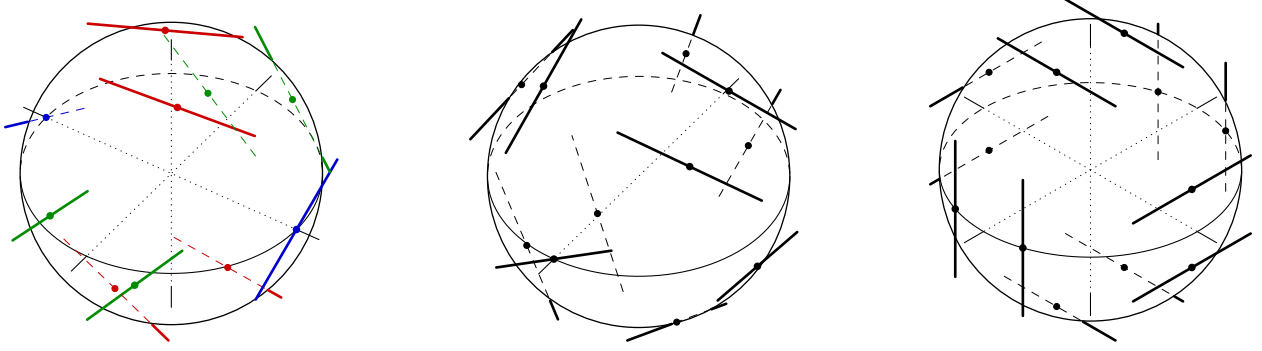
For each n in the window left open by Theorems 1.5 and 1.8, except $n = 8$, we give $n + 1$ tangent lines with rational data. Their pairwise distances are computed exactly and compared with a rational upper bound for $2 \sin \frac{\pi}{n}$. The pinwheel is not used in this section.

Theorem 6.1. Let $W = \{9, 10, \dots, 18\} \cup \{20, 22, \dots, 36\}$. For every $n \in W$,

$$N(\omega_n) \geq n + 1, \quad \text{i.e.} \quad (n, \omega_n) \notin \mathcal{P}_{\text{cyl}}.$$

Proof of Theorem 1.6. Theorem 6.1 covers W , and Theorem 1.8 covers every odd $n \geq 19$ and every even $n \geq 38$. The union is $\{n \geq 9\}$. $\square$

Together with Theorem 1.5, the insertion question (Question 1.1) is decided for every $n \geq 3$ except $n = 8$: the ring of n is count-optimal at its own radius for $3 \leq n \leq 7$ and is not for any $n \geq 9$.

FIGURE 4. The configurations that show $(9, \omega_9)$, $(10, \omega_{10})$ and $(11, \omega_{11}) \notin \mathcal{P}_{\text{cyl}}$.

6.1. Rational tangent lines. The witnesses are lines tangent to $\mathbb{S}^2$ at rational points, with integer directions. Normalizing the direction is unnecessary, because the distance formula of Lemma 2.3 is homogeneous of degree zero in each direction vector.

Definition 6.2. A rational tangent line is a pair $(P, \vec{V})$ of integer vectors $P = (A, B, C, E) \in \mathbb{Z}^4$ and $\vec{V} = (V_0, V_1, V_2) \in \mathbb{Z}^3 \setminus \{0\}$ with

$$(W1) \quad A^2 + B^2 + C^2 = E^2, \quad E > 0, \quad (W2) \quad AV_0 + BV_1 + CV_2 = 0, \quad \vec{V} \neq 0. \quad (27)$$

It denotes the line $\ell(P, \vec{V}) = \{\vec{p} + t\vec{V} : t \in \mathbb{R}\}$ with $\vec{p} = \frac{1}{E}(A, B, C)$, which by (W1) and (W2) is tangent to the unit sphere at $\vec{p}$.

The following result follows from Lemma 2.3. In the parallel case we also use the identity $\|\vec{x} \times \vec{y}\|^2 = \|\vec{x}\|^2 \|\vec{y}\|^2 - (\vec{x} \cdot \vec{y})^2$ for $\vec{x}, \vec{y} \in \mathbb{R}^3$.

Lemma 6.3. Let $\ell = \ell(P, \vec{V})$ and $\ell' = \ell(P', \vec{V}')$ be rational tangent lines with tangency points $\vec{p}, \vec{p}'$. Define $\vec{c} = \vec{V} \times \vec{V}'$ and $\vec{\Delta} = \vec{p}' - \vec{p}$. Then $\text{dist}(\ell, \ell')^2$ is the rational number

$$\frac{(\vec{c} \cdot \vec{\Delta})^2}{\|\vec{c}\|^2} \quad \text{if } \vec{c} \neq 0, \quad \left\| \vec{\Delta} - \frac{\vec{\Delta} \cdot \vec{V}}{\|\vec{V}\|^2} \vec{V} \right\|^2 \quad \text{if } \vec{c} = 0. \quad (28)$$

The threshold $2 \sin \frac{\pi}{n}$ is irrational. We replace it by a slightly larger rational number.

Lemma 6.4. The inequality $4 \sin^2 \frac{\pi}{n} \leq r_n$ holds for every integer $n \geq 1$, where the rational numbers Y_n and r_n are defined as $Y_n = \frac{710}{113n}$ and $r_n = Y_n^2 - \frac{Y_n^4}{12} + \frac{Y_n^6}{360}$.

Proof. For $y \geq 0$,

$$\cos y \geq 1 - \frac{y^2}{2} + \frac{y^4}{24} - \frac{y^6}{720}. \quad (29)$$

Indeed, the difference of the two sides vanishes at 0, its derivative $-\sin y + y - \frac{y^3}{6} + \frac{y^5}{120}$ vanishes at 0, the derivative of that, $-\cos y + 1 - \frac{y^2}{2} + \frac{y^4}{24}$, vanishes at 0, and its derivative $\sin y - y + \frac{y^3}{6}$ is nonnegative by (E1). Integrating three times from 0 gives (29). Hence $4 \sin^2 \frac{\pi}{n} = 2(1 - \cos \frac{2\pi}{n}) \leq \rho(2\pi/n)$ with $\rho(y) = y^2 - \frac{y^4}{12} + \frac{y^6}{360}$. Now $\rho'(y) = \frac{y}{60}(y^4 - 20y^2 + 120) > 0$ for $y > 0$ (the quadratic in y^2 has negative discriminant), so ρ is increasing on $[0, \infty)$, and since $\pi < \frac{355}{113}$ we have $\frac{2\pi}{n} < Y_n$ and $\rho(2\pi/n) \leq \rho(Y_n) = r_n$. $\square$

6.2. The deposited configurations and their verification. For each $N \in \{10, 11, 12, \dots, 21\} \cup \{23, 25, \dots, 37\}$ a file

cyl_window_N{N}.txt

contains a configuration that, for $n = N - 1$, proves $(n, \omega_n) \notin \mathcal{P}_{\text{cyl}}$. The file contains N rows of seven integers $A B C E V_0 V_1 V_2$. Each row corresponds to one rational tangent line. All integers have at most 13 decimal digits. The verifier (Section 5.4) performs the following checks:

- (W1) for every row, $A^2 + B^2 + C^2 = E^2$ and $E > 0$;
(W2) for every row, $AV_0 + BV_1 + CV_2 = 0$ and $(V_0, V_1, V_2) \neq (0, 0, 0)$;
(W3) for every pair of rows, the rational squared distance of Lemma 6.3 is at least r_n .

Proof of Theorem 6.1. Let $n \in W$ and $N = n + 1$. The deposited file with N rows passes (W1)–(W3). By (W1) and (W2) the rows are N lines tangent to $\mathbb{S}^2$. By Lemma 6.3 and (W3) their pairwise squared distances are at least r_n , and by Lemma 6.4 $r_n \geq 4 \sin^2 \frac{\pi}{n} = \tau(\omega_n)^2$ (Remark 2.2). So N tangent lines have pairwise distances at least $\tau(\omega_n)$, and Lemma 2.1 gives $N = n + 1$ non-overlapping cylinders of radius ω_n touching B . $\square$

7. THE TWO-POINT BOUND

The bound we use is the two-point bound for packing problems on a compact group. For the rotation group it was first worked out by de Oliveira Filho, Spomer, and Vallentin [25], in the framework of de Laat and Vallentin [11]. We state and prove it in the form we need.

Throughout this section, $\tau \in (0, 2)$ is fixed, $L \in \mathbb{N}$, and, for $\ell = 1, \dots, L$, we fix

- (i) a symmetric positive-definite matrix $D_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, and
- (ii) a map $T^\ell : \text{SO}(3) \rightarrow \mathbb{R}^{n_\ell \times n_\ell}$ satisfying, for all $g_1, g_2, g \in \text{SO}(3)$,

$$T^\ell(g_1 g_2) = T^\ell(g_1) T^\ell(g_2) \quad \text{and} \quad T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell.$$

Lemma 7.1. *Under (i)–(ii), for every $g \in \text{SO}(3)$: $T^\ell(e) = I$ and*

$$T^\ell(g^{-1}) = T^\ell(g)^{-1} = D_\ell^{-1} T^\ell(g)^\top D_\ell.$$

Proof. Since D_ℓ is invertible, the identity $T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell$ gives $(D_\ell^{-1} T^\ell(g)^\top D_\ell) T^\ell(g) = I$. So $T^\ell(g)$ is invertible and $T^\ell(g)^{-1} = D_\ell^{-1} T^\ell(g)^\top D_\ell$. Now $T^\ell(e) = T^\ell(e \cdot e) = T^\ell(e)^2$ with $T^\ell(e)$ invertible forces $T^\ell(e) = I$. Finally $T^\ell(g) T^\ell(g^{-1}) = T^\ell(e) = I$ gives $T^\ell(g^{-1}) = T^\ell(g)^{-1}$. $\square$

Definition 7.2. For a sequence $\vec{C} = (C_\ell)_{\ell=1}^L$ of symmetric positive-semidefinite matrices $C_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$, we define its supremum over the safe zone as

$$S(\vec{C}) = \sup_{g \in \mathcal{D}(\tau)} \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell T^\ell(g)).$$

Remark 7.3. When $D_\ell = \text{diag}(w_1^\ell, \dots, w_{n_\ell}^\ell)$ is diagonal and C_ℓ is symmetric,

$$\text{tr}(D_\ell^{-1} C_\ell T^\ell(g)) = \sum_{i,j} \frac{(C_\ell)_{ij}}{w_j^\ell} T^\ell(g)_{ij},$$

the sum of the entries of the Schur product of C_ℓ and $T^\ell(g)$, with the j -th column weighted by $1/w_j^\ell$. In particular $\text{tr}(D_\ell^{-1} C_\ell T^\ell(e)) = \text{tr}(D_\ell^{-1} C_\ell) = \sum_i (C_\ell)_{ii}/w_i^\ell$.

Theorem 7.4. *Let $\vec{C} = (C_\ell)_{\ell=1}^L$ be a sequence of symmetric positive-semidefinite matrices $C_\ell \in \mathbb{R}^{n_\ell \times n_\ell}$. If $S(\vec{C}) \leq -1$, then*

$$N_{\max}(\tau) \leq 1 + \left\lfloor \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell) \right\rfloor. \quad (30)$$

Proof. Let $g_1, \dots, g_N$ be distinct elements of a non-overlapping set (Definition 2.5), and put $S_\ell = \sum_{i=1}^N T^\ell(g_i)$. By Lemma 7.1,

$$T^\ell(g_i^{-1} g_j) = T^\ell(g_i)^{-1} T^\ell(g_j) = D_\ell^{-1} T^\ell(g_i)^\top D_\ell T^\ell(g_j),$$

so that, summing over all ordered pairs and using linearity and cyclicity of the trace,

$$\begin{aligned} \sum_{i=1}^N \sum_{j=1}^N \text{tr}(D_\ell^{-1} C_\ell T^\ell(g_i^{-1} g_j)) &= \sum_{i,j} \text{tr}(D_\ell^{-1} C_\ell D_\ell^{-1} T^\ell(g_i)^\top D_\ell T^\ell(g_j)) \\ &= \text{tr}\left((D_\ell^{-1} C_\ell D_\ell^{-1})(S_\ell^\top D_\ell S_\ell)\right) \geq 0, \end{aligned}$$

since $D_\ell^{-1} C_\ell D_\ell^{-1}$ and $S_\ell^\top D_\ell S_\ell$ are congruent to C_ℓ and D_ℓ , hence symmetric positive semidefinite, and for such P, Q we have $\text{tr}(PQ) = \text{tr}(\sqrt{Q} P \sqrt{Q}) \geq 0$.

Summing over ℓ and separating the N diagonal pairs $i = j$ (where $T^\ell(e) = I$ gives the value $\sum_\ell \text{tr}(D_\ell^{-1}C_\ell)$) from the $N(N-1)$ off-diagonal pairs (where $g_i^{-1}g_j \in \mathcal{D}$ by Definition 2.5, so each term is at most $S(\vec{C}) \leq -1$),

$$0 \leq N \sum_{\ell=1}^L \text{tr}(D_\ell^{-1}C_\ell) - N(N-1).$$

Dividing by N gives $N-1 \leq \sum_\ell \text{tr}(D_\ell^{-1}C_\ell)$, which implies (30). $\square$

Remark 7.5. In the deposited certificates $T_h^\ell(q)$ is the matrix of the action of $R(q)$ on the harmonic polynomials of degree ℓ on $\mathbb{R}^3$, in an integer orthogonal basis. The tables are computed as in [16]. These actions are the irreducible representations of $\text{SO}(3)$, one in each dimension $2\ell+1$ [2]. By Bochner's theorem [1], in the form [25], every continuous function of positive type on $\text{SO}(3)$ has the form $\sum_{\ell \geq 0} \text{tr}(F_\ell T^\ell(g))$ with positive semidefinite matrices F_ℓ . The constant -1 in Theorem 7.4 plays the role of the term $\ell = 0$. So the bound uses all such functions of degree at most L . It is the Lovász theta number [13] of the packing graph on $\text{SO}(3)$, truncated at degree L .

8. REDUCTION TO A POLYNOMIAL PROBLEM

We will now find an equivalent problem in which $\text{SO}(3)$ does not appear, and whose only objects are polynomials in the four variables q_0, q_1, q_2 , and q_3 .

8.1. Quaternions and rotations. For $q = (q_0, q_1, q_2, q_3) \in \mathbb{R}^4$ define

$$R(q) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix},$$

whose entries are integer quadratic forms in q , and write $|q|^2 = q_0^2 + q_1^2 + q_2^2 + q_3^2$. Let $p * q$ denote the Hamilton product,

$$\begin{aligned} (p * q)_0 &= p_0q_0 - p_1q_1 - p_2q_2 - p_3q_3, \\ (p * q)_1 &= p_0q_1 + q_0p_1 + p_2q_3 - p_3q_2, \\ (p * q)_2 &= p_0q_2 + q_0p_2 + p_3q_1 - p_1q_3, \\ (p * q)_3 &= p_0q_3 + q_0p_3 + p_1q_2 - p_2q_1. \end{aligned}$$

Lemma 8.1. *The following hold as polynomial identities in $\mathbb{Z}[p, q]$:*

- (a) $R(q)^\top R(q) = |q|^4 I$ and $\det R(q) = |q|^6$. In particular, $R(q) \in \text{SO}(3)$ whenever $|q| = 1$;
- (b) $R(p * q) = R(p) R(q)$ and $|p * q|^2 = |p|^2 |q|^2$;
- (c) the ten products $q_a q_b$ ($0 \leq a \leq b \leq 3$) are $\mathbb{Z}$ -linear combinations of $|q|^2$ and the entries R_{ij} of $R(q)$. Explicitly, $4q_0^2 = |q|^2 + R_{11} + R_{22} + R_{33}$, $4q_1^2 = |q|^2 + R_{11} - R_{22} - R_{33}$, $4q_2^2 = |q|^2 - R_{11} + R_{22} - R_{33}$, $4q_3^2 = |q|^2 - R_{11} - R_{22} + R_{33}$, $4q_0q_1 = R_{32} - R_{23}$, $4q_0q_2 = R_{13} - R_{31}$, $4q_0q_3 = R_{21} - R_{12}$, $4q_1q_2 = R_{12} + R_{21}$, $4q_1q_3 = R_{13} + R_{31}$, $4q_2q_3 = R_{23} + R_{32}$.

Proof. Each identity is an equality of polynomials with integer coefficients, of degree at most 6 in q in (a) and (c) and at most 4 in p, q in (b), and is verified by expanding both sides. $\square$

Lemma 8.2.

- (a) If q, q' are unit quaternions with $R(q) = R(q')$, then $q' = \pm q$.
- (b) Every $g \in \text{SO}(3)$ equals $R(q)$ for some unit quaternion q .

Proof. (a) By Lemma 8.1(c) the entries of $R(q)$ together with $|q|^2 = 1$ determine every product $q_a q_b$, i.e. the symmetric rank-one matrix $qq^\top$. If $qq^\top = q'q'^\top$ then, taking any index a with $q_a \neq 0$ (such an index exists because $|q| = 1$), the a -th columns give $q_a q = q'_a q'$ with $q_a^2 = (q'_a)^2$, so $q' = \pm q$.

(b) First, every $g \in \text{SO}(3)$ has an eigenvector corresponding to eigenvalue 1. Indeed, since $\det g = 1$ and $g^\top g = I$,

$$\det(g - I) = \det(g) \det(I - g^\top) = \det((I - g)^\top) = \det(I - g) = (-1)^3 \det(g - I).$$

Therefore, $\det(g - I) = 0$ and there is a unit vector $\vec{n}$ with $g\vec{n} = \vec{n}$. Since g is orthogonal and fixes $\vec{n}$, it maps the plane $\vec{n}^\perp$ to itself, and its restriction there is an orientation-preserving linear isometry of a plane, hence a rotation by some angle θ . Let $\vec{n} = (n_x, n_y, n_z)$ and set

$$q = \left(\cos \frac{\theta}{2}, n_x \sin \frac{\theta}{2}, n_y \sin \frac{\theta}{2}, n_z \sin \frac{\theta}{2} \right).$$

Clearly, q is a unit quaternion. A direct computation from the definition of $R(q)$ shows $R(q)\vec{n} = \vec{n}$ and that $R(q)$ rotates $\vec{n}^\perp$ by θ with the same orientation. Therefore, $R(q) = g$. $\square$

8.2. The safe zone as a pair of polynomial inequalities. By Lemma 8.2 the map $q \mapsto R(q)$ carries the unit sphere of $\mathbb{R}^4$ onto $\text{SO}(3)$, two-to-one. We now describe the preimage of the safe zone $\mathcal{D}(\tau)$ of Definition 2.4 by two polynomial inequalities in q .

Lemma 8.3. *Let $q \neq 0$ and $g = |q|^{-2}R(q)$. Then*

$$\|\vec{e}_1 \times g\vec{e}_1\|^2 = \frac{4(q_0^2 + q_1^2)(q_2^2 + q_3^2)}{|q|^4}, \quad (31)$$

$$(g\vec{e}_3 - \vec{e}_3) \cdot (\vec{e}_1 \times g\vec{e}_1) = \frac{-4q_1q_2}{|q|^2}, \quad (32)$$

$$\|(g\vec{e}_3 - \vec{e}_3) \times \vec{e}_1\|^2 = \frac{4[(q_2q_3 - q_0q_1)^2 + (q_1^2 + q_2^2)^2]}{|q|^4}. \quad (33)$$

Proof. Write R_{ij} for the entries of $R(q)$. The vectors $g\vec{e}_1$ and $g\vec{e}_3$ are the first and the third column of g . Hence

$$\begin{aligned} \vec{e}_1 \times g\vec{e}_1 &= |q|^{-2}(0, -R_{31}, R_{21}), \\ g\vec{e}_3 - \vec{e}_3 &= |q|^{-2}(R_{13}, R_{23}, R_{33} - |q|^2), \end{aligned}$$

and $(g\vec{e}_3 - \vec{e}_3) \times \vec{e}_1 = |q|^{-2}(0, R_{33} - |q|^2, -R_{23})$. From the definition of $R(q)$ in Section 8.1,

$$\begin{aligned} R_{21} &= 2(q_1q_2 + q_0q_3), & R_{23} &= 2(q_2q_3 - q_0q_1), \\ R_{31} &= 2(q_1q_3 - q_0q_2), & R_{33} - |q|^2 &= -2(q_1^2 + q_2^2). \end{aligned}$$

Substituting these gives (31) and (33) at once. In (31) the cross terms $\pm 8q_0q_1q_2q_3$ in $R_{21}^2 + R_{31}^2$ cancel. For (32), the left-hand side is $|q|^{-4}(-R_{23}R_{31} + (R_{33} - |q|^2)R_{21})$, and

$$\begin{aligned} &-R_{23}R_{31} + (R_{33} - |q|^2)R_{21} \\ &= -4(q_2q_3 - q_0q_1)(q_1q_3 - q_0q_2) - 4(q_1^2 + q_2^2)(q_1q_2 + q_0q_3) \\ &= -4(q_1q_2q_3^2 - q_0q_2^2q_3 - q_0q_1^2q_3 + q_0^2q_1q_2) \\ &\quad - 4(q_1^3q_2 + q_0q_1^2q_3 + q_1q_2^3 + q_0q_2^2q_3) \\ &= -4q_1q_2(q_0^2 + q_1^2 + q_2^2 + q_3^2). \end{aligned}$$

The four terms carrying q_0q_3 cancel in pairs. $\square$

Write $\tau^2 = A/B$ with A, B positive coprime integers and set

$$Q_h(q) = 4B q_1^2 q_2^2 - A(q_0^2 + q_1^2)(q_2^2 + q_3^2), \quad (34)$$

$$U_h(q) = 4B[(q_2q_3 - q_0q_1)^2 + (q_1^2 + q_2^2)^2] - A(|q|^2)^2. \quad (35)$$

These are homogeneous of degree 4 with integer coefficients, and both are invariant under $q \mapsto -q$.

Theorem 8.4. *For every $q \neq 0$,*

$$|q|^{-2}R(q) \in \mathcal{D}(\tau) \iff Q_h(q) \geq 0 \text{ and } U_h(q) \geq 0. \quad (36)$$

Proof. Let $g = |q|^{-2}R(q)$ and $\delta = \text{dist}(\ell_0, g \cdot \ell_0)$. The line $g \cdot \ell_0$ passes through $g\vec{e}_3$ with direction $g\vec{e}_1$. Let d be the distance from the point $g\vec{e}_3$ to the line ℓ_0 . Since $\vec{e}_1$ is a unit vector, $d = \|(g\vec{e}_3 - \vec{e}_3) \times \vec{e}_1\|$. By (33) and (35), $U_h(q) = B|q|^4(d^2 - \tau^2)$. So $U_h(q) \geq 0$ holds exactly when $d \geq \tau$. Also $\delta \leq d$, because $g\vec{e}_3$ lies on $g \cdot \ell_0$.

Case $(q_0^2 + q_1^2)(q_2^2 + q_3^2) > 0$. By (31), $\vec{e}_1 \times g\vec{e}_1 \neq 0$, so the lines are not parallel. Lemma 2.3 with $\vec{p}_1 = \vec{e}_3$, $\vec{v}_1 = \vec{e}_1$, $\vec{p}_2 = g\vec{e}_3$ and $\vec{v}_2 = g\vec{e}_1$, together with (31) and (32), gives

$$\delta^2 = \frac{4q_1^2 q_2^2}{(q_0^2 + q_1^2)(q_2^2 + q_3^2)}.$$

Hence $Q_h(q) = B(q_0^2 + q_1^2)(q_2^2 + q_3^2)(\delta^2 - \tau^2)$. So $Q_h(q) \geq 0$ holds exactly when $\delta \geq \tau$. In that case $d \geq \delta \geq \tau$, so $U_h(q) \geq 0$ automatically.

Case $(q_0^2 + q_1^2)(q_2^2 + q_3^2) = 0$. Then $q_0 = q_1 = 0$ or $q_2 = q_3 = 0$. In both cases $q_1 q_2 = 0$, so $Q_h(q) = 0$. By (31) the lines are parallel. So $\delta = d$ by Lemma 2.3, and $\delta \geq \tau$ holds exactly when $U_h(q) \geq 0$. $\square$

We write

$$\mathcal{D}_h(\tau) = \{q \in \mathbb{R}^4 \setminus \{0\} : Q_h(q) \geq 0, U_h(q) \geq 0\},$$

a cone that is invariant under $q \mapsto \lambda q$ for every $\lambda \neq 0$. By Lemma 8.2, $\mathcal{D}_h(\tau)$ is the full preimage of $\mathcal{D}(\tau)$ under $q \mapsto |q|^{-2}R(q)$.

9. THE CERTIFICATE THEOREM

We can now state the general result. Hypotheses (H1)–(H5) below are finite assertions about rational numbers and polynomials with rational coefficients, verified by inspection, expansion, or an explicit positivity witness. Hypothesis (H6) is an inequality on a semialgebraic set. Section 12 proves it by a sum-of-squares identity.

Definition 9.1. Let $\tau \in (0, 2)$ with $\tau^2 \in \mathbb{Q}$, let $L \in \mathbb{N}$, and let $T^* \in \mathbb{Q}$. A *certificate at (τ, L, T^*)* consists of, for each $\ell = 1, \dots, L$:

- a diagonal matrix $D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell)$ with $w_i^\ell \in \mathbb{Q}_{>0}$,
- a symmetric matrix $C_\ell \in \mathbb{Q}^{(2\ell+1) \times (2\ell+1)}$, and
- a matrix T_h^ℓ of polynomials in $\mathbb{Q}[q_0, q_1, q_2, q_3]$ of size $(2\ell+1) \times (2\ell+1)$,

subject to the following hypotheses. Set

$$P(q) = \sum_{\ell=1}^L (|q|^2)^{L-\ell} \sum_{i,j=1}^{2\ell+1} \frac{(C_\ell)_{ij} (T_h^\ell)_{ij}(q)}{w_j^\ell}, \quad (37)$$

$$G(q) = -(|q|^2)^L - P(q). \quad (38)$$

(H1) (*Homogeneity.*) Every entry of T_h^ℓ is a homogeneous polynomial of degree 2ℓ .

(H2) (*Multiplicativity.*) $T_h^\ell(p * q) = T_h^\ell(p) T_h^\ell(q)$ as an identity in $\mathbb{Q}[p, q]$.

(H3) (*Orthogonality.*) $T_h^\ell(q)^\top D_\ell T_h^\ell(q) = (|q|^2)^{2\ell} D_\ell$ as an identity in $\mathbb{Q}[q]$.

(H4) (*Positive semidefiniteness.*) C_ℓ is positive semidefinite.

(H5) (*Trace.*) $\sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell) = T^*$.

(H6) (*Cover.*) $G(q) \geq 0$ for every $q \in \mathcal{D}_h(\tau)$.

Theorem 9.2. If a certificate at (τ, L, T^*) exists, then

$$N_{\max}(\tau) \leq 1 + \lfloor T^* \rfloor.$$

Proof. Step 1: T_h^ℓ descends to a map on $\text{SO}(3)$. For $g \in \text{SO}(3)$ choose, by Lemma 8.2(b), a unit quaternion q with $R(q) = g$, and set $T^\ell(g) = T_h^\ell(q)$. This is well defined: by Lemma 8.2(a) the only other choice is $-q$, and by (H1) the entries of T_h^ℓ are homogeneous of the even degree 2ℓ , so $T_h^\ell(-q) = T_h^\ell(q)$.

Step 2: the hypotheses of Section 7. Each D_ℓ is diagonal with positive entries, hence symmetric positive definite. Let $g_1, g_2 \in \text{SO}(3)$ and pick unit quaternions $q^{(1)}, q^{(2)}$ with $R(q^{(i)}) = g_i$. By Lemma 8.1(b), $R(q^{(1)} * q^{(2)}) = g_1 g_2$ and $|q^{(1)} * q^{(2)}| = 1$, so (H2) gives

$$T^\ell(g_1 g_2) = T_h^\ell(q^{(1)} * q^{(2)}) = T_h^\ell(q^{(1)}) T_h^\ell(q^{(2)}) = T^\ell(g_1) T^\ell(g_2),$$

and (H3) at a unit quaternion gives $T^\ell(g)^\top D_\ell T^\ell(g) = D_\ell$. Thus the data $(D_\ell, T^\ell)_{\ell=1}^L$ satisfy (i)–(ii) of Section 7, and by (H4) the sequence $\vec{C} = (C_\ell)_{\ell=1}^L$ is admissible in Definition 7.2.

certificate	τ	$\omega = \frac{\tau}{2-\tau}$	L	proves
cyl_N4_omega39_25	$\frac{39}{32}$	$\frac{39}{25}$	4	$(4, \frac{39}{25}) \in \mathcal{P}_{\text{cyl}}$
cyl_N5_omega1203_845	$\frac{1203}{1024}$	$\frac{1203}{845}$	4	$(5, \frac{1203}{845}) \in \mathcal{P}_{\text{cyl}}$
cyl_N6_omega1_1	1	1	4	$(6, 1) \in \mathcal{P}_{\text{cyl}}$
cyl_N7_omega14558721_18995711	$\frac{14558721}{2^{24}}$	$\frac{14558721}{18995711}$	6	$(7, \frac{14558721}{18995711}) \in \mathcal{P}_{\text{cyl}}$

TABLE 2. The deposited certificates. Each file is named after the statement it proves.

Step 3: P computes the two-point sum. For a unit quaternion q we have $|q|^2 = 1$, so by (37) and Remark 7.3,

$$P(q) = \sum_{\ell=1}^L \sum_{i,j} \frac{(C_\ell)_{ij} (T_h^\ell)_{ij}(q)}{w_j^\ell} = \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell T^\ell(R(q))).$$

By (H1) the ℓ -th summand of (37) is homogeneous of degree $2(L - \ell) + 2\ell = 2L$, so P is homogeneous of degree $2L$. Hence for arbitrary $q \neq 0$,

$$\frac{P(q)}{(|q|^2)^L} = \sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell T^\ell(|q|^{-2} R(q))). \quad (39)$$

Step 4: (H6) is exactly $S(\vec{C}) \leq -1$. Let $g \in \mathcal{D}(\tau)$ and pick $q \neq 0$ with $g = |q|^{-2} R(q)$. By (36) we have $q \in \mathcal{D}_h(\tau)$, so (H6) gives $G(q) \geq 0$, i.e. $P(q) \leq -(|q|^2)^L$. Dividing by $(|q|^2)^L > 0$ and using (39),

$$\sum_{\ell=1}^L \text{tr}(D_\ell^{-1} C_\ell T^\ell(g)) \leq -1.$$

Taking the supremum over $g \in \mathcal{D}(\tau)$ yields $S(\vec{C}) \leq -1$.

Step 5: conclusion. By (H5) and Theorem 7.4,

$$N_{\max}(\tau) \leq 1 + \left\lfloor \sum_{\ell} \text{tr}(D_\ell^{-1} C_\ell) \right\rfloor = 1 + \lfloor T^* \rfloor \square$$

In the language of Definition 1.3, the conclusion of Theorem 9.2 reads $(1 + \lfloor T^* \rfloor, \tau/(2 - \tau)) \in \mathcal{P}_{\text{cyl}}$.

10. NO INSERTION FOR $3 \leq n \leq 7$: PROOF OF THEOREM 1.5

The four certificates of Table 2 prove the cases $4 \leq n \leq 7$ of Theorem 1.5. They share the matrices D_ℓ and T_h^ℓ of Section 11. They differ in τ^2 , L , the matrices C_ℓ , the number T^* , and the sum-of-squares data of Section 12.

By Lemma 1.4 it suffices to prove $(n, \omega') \in \mathcal{P}_{\text{cyl}}$ for some rational $\omega' \leq \omega_n$. The values of ω' in Table 2 are simply the ones our search produced.

Proof of Theorem 1.5. $n = 3$. We have $\omega_3 = 3 + 2\sqrt{3} > 1 + \sqrt{2}$. Theorem 1.7 gives $(3, \omega_3) \in \mathcal{P}_{\text{cyl}}$.

$n = 4$. The certificate cyl_N4_omega39_25, at $\tau = \frac{39}{32}$, gives $N_{\max}(\frac{39}{32}) \leq 4$ by Theorem 9.2. So $(4, \frac{39}{25}) \in \mathcal{P}_{\text{cyl}}$. Also $\frac{39}{25} \leq 1 + \sqrt{2} = \omega_4$, since $14^2 = 196 \leq 1250 = 2 \cdot 25^2$.

$n = 5$. The certificate cyl_N5_omega1203_845, at $\tau = \frac{1203}{1024}$, gives $N_{\max}(\frac{1203}{1024}) \leq 5$. So $(5, \frac{1203}{845}) \in \mathcal{P}_{\text{cyl}}$. It remains to check $\omega_5 \geq \frac{1203}{845}$, that is, $\sin \frac{\pi}{5} \geq \frac{1203}{2048}$. Squaring and using $16 \sin^2 \frac{\pi}{5} = 10 - 2\sqrt{5}$, this is equivalent in turn to

$$10 - 2\sqrt{5} \geq \frac{1447209}{262144} \iff 1174231 \geq 524288\sqrt{5} \iff 1174231^2 \geq 5 \cdot 524288^2.$$

Indeed $1378818441361 \geq 1374389534720$.

$n = 6$. The certificate cyl_N6_omega1_1 gives $N_{\max}(1) \leq 6$. This was first proved in [16]. Since $2 \sin \frac{\pi}{6} = 1$ corresponds to $\omega = \omega_6 = 1$, we get $(6, \omega_6) \in \mathcal{P}_{\text{cyl}}$.

$n = 7$. The certificate cyl_N7_omega14558721_18995711, at $\tau = \frac{14558721}{2^{24}}$, gives $N_{\max}(\tau) \leq 7$. So $(7, \frac{\tau}{2-\tau}) \in \mathcal{P}_{\text{cyl}}$. It remains to check $\omega_7 \geq \frac{\tau}{2-\tau}$, that is, $\sin \frac{\pi}{7} \geq \frac{\tau}{2}$. We have $\sin^2 \frac{\pi}{7} = \frac{1}{2}(1 - c)$, where $c = \cos \frac{2\pi}{7}$ is the largest root of $p(x) = 8x^3 + 4x^2 - 4x - 1$. So the inequality is equivalent to $c \leq r$, where $r = 1 - \frac{\tau^2}{2}$. The polynomial p is increasing

to the right of its largest root, and $r > 0.62$ exceeds the other two roots. Hence the inequality is equivalent to $p(r) \geq 0$. With $\tau^2 = \frac{211956357155841}{2^{48}}$, the inequality $p(r) > 0$ is a comparison of two integers, which holds. $\square$

Remark 10.1. The case $n = 4$ of Theorem 1.5 was proved earlier. Ogievetsky and Shlosman [18, Section 6.3] proved it with an argument about inscribed balls. Firsching [7], building on Braß and Wenk [5], gave a stronger bound.

11. THE CERTIFICATE DATA

This section describes the deposited data and the checks that establish (H1)–(H5). How the data was found plays no role here. The *universal* part, shared by all certificates, consists, for each $\ell = 1, \dots, 6$, of:

- positive rationals $w_1^\ell, \dots, w_{2\ell+1}^\ell$, forming the diagonal matrix $D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell)$;
- an integer table defining a $(2\ell + 1) \times (2\ell + 1)$ matrix $T_h^\ell(q)$ of polynomials in $q = (q_0, q_1, q_2, q_3)$, with coefficients in $\frac{1}{\text{Tden}_\ell} \mathbb{Z}$ for a deposited scale $\text{Tden}_\ell \in \{1, 2, 4\}$ (the deposited values are 1, 1, 2, 1, 4, 2 for $\ell = 1, \dots, 6$).

A certificate with truncation L uses the first L universal blocks. The deposited certificates have $L = 4$ or $L = 6$. Each certificate additionally deposits the rational τ^2 , the number T^* with a scale M (Appendix B), and symmetric rational matrices $C_\ell \in \mathbb{Q}^{(2\ell+1) \times (2\ell+1)}$ together with integer witnesses for their positive semidefiniteness.

Positivity of each w_i^ℓ , required by Definition 9.1, is read off from the deposited numerators and denominators. Hypotheses (H1), (H2), and (H3) are finite computations on the universal tables, performed once for all certificates. Hypothesis (H4) follows from the deposited witnesses by Proposition 11.2, and (H5) is a finite computation per certificate. Hypothesis (H6) is the subject of Section 12.

11.1. The identities (H1)–(H3). Hypothesis (H1) is an inspection of the exponents in the deposited table: every row of T_h^ℓ carries exponents summing to 2ℓ .

Both (H2) and (H3) are identities between polynomials with integer coefficients once the deposited scales are cleared. Writing $A^\ell = \text{Tden}_\ell T_h^\ell$, which has integer entries, and $\widehat{D}_\ell = m_\ell D_\ell$, where m_ℓ is a common denominator of $w_1^\ell, \dots, w_{2\ell+1}^\ell$, they read

$$\text{Tden}_\ell A^\ell (p * q) = A^\ell(p) A^\ell(q), \quad A^\ell(q)^\top \widehat{D}_\ell A^\ell(q) = \text{Tden}_\ell^2 |q|^{4\ell} \widehat{D}_\ell,$$

so each amounts to comparing two polynomial expansions with integer coefficients, coefficient by coefficient.

11.2. Positive semidefiniteness (H4). The positive semidefiniteness of the matrices $C_1, \dots, C_L$ of each certificate follows from the deposited integer witnesses by the following two elementary facts. The first is a special case of Gershgorin's circle theorem [8], which states that every eigenvalue λ of a square matrix E satisfies $|\lambda - E_{ii}| \leq \sum_{j \neq i} |E_{ij}|$ for some i .

Lemma 11.1. *Let $E \in \mathbb{R}^{n \times n}$ be symmetric with $E_{ii} \geq \sum_{j \neq i} |E_{ij}|$ for every i . Then $E \succeq 0$.*

Proof. Let $v \in \mathbb{R}^n$. Using $|v_i v_j| \leq \frac{1}{2}(v_i^2 + v_j^2)$ and then the symmetry of E ,

$$\left| \sum_{i \neq j} E_{ij} v_i v_j \right| \leq \sum_{i \neq j} |E_{ij}| \frac{1}{2}(v_i^2 + v_j^2) = \sum_i v_i^2 \sum_{j \neq i} |E_{ij}| \leq \sum_i E_{ii} v_i^2,$$

so $v^\top E v = \sum_i E_{ii} v_i^2 + \sum_{i \neq j} E_{ij} v_i v_j \geq 0$. $\square$

Proposition 11.2. *Write $\widehat{C}_\ell = 2^\kappa C_\ell$ for the deposited integer matrix, where κ is the even integer deposited as cDenLog2 , and let S_ℓ and W_ℓ be the deposited positive integer scalar and integer matrix. If*

$$E_\ell = S_\ell \widehat{C}_\ell - W_\ell^\top W_\ell$$

is symmetric and diagonally dominant, in the sense that $(E_\ell)_{ii} \geq \sum_{j \neq i} |(E_\ell)_{ij}|$ for every i , then C_ℓ is symmetric positive semidefinite.

Proof. By Lemma 11.1, $E_\ell \succeq 0$. Also $W_\ell^\top W_\ell \succeq 0$, since $v^\top W_\ell^\top W_\ell v = \|W_\ell v\|^2 \geq 0$ for every v . Hence

$$2^\kappa S_\ell C_\ell = S_\ell \widehat{C}_\ell = E_\ell + W_\ell^\top W_\ell \succeq 0,$$

and dividing by the positive scalar $2^\kappa S_\ell$ gives $C_\ell \succeq 0$. Symmetry of C_ℓ is symmetry of the deposited $\widehat{C}_\ell$. $\square$

11.3. The trace sum (H5). The verifier evaluates $\sum_\ell \sum_i (C_\ell)_{ii} / w_i^\ell$ exactly and checks that it equals the deposited T^* . This is (H5).

11.4. The certificate polynomial. The polynomial P of Definition 9.1, formed from the deposited C_ℓ , the tables T_h^ℓ and the weights, has rational coefficients and is homogeneous of degree $2L$: by (H1) each $T_h^\ell(q)_{ij}$ is homogeneous of degree 2ℓ , and $2(L - \ell) + 2\ell = 2L$ for every ℓ . The same holds for $G = -(|q|^2)^L - P$.

12. A SUM-OF-SQUARES PROOF OF HYPOTHESIS (H6)

By Theorem 9.2 and Section 11, each certificate proves its bound once (H6) is established for it, that is, once we show

$$G(q) \geq 0 \quad \text{on} \quad \mathcal{D}_h(\tau) = \{q \neq 0 : Q_h(q) \geq 0 \text{ and } U_h(q) \geq 0\}, \quad (40)$$

with Q_h, U_h formed from the certificate's $\tau^2 = A/B$ by (34)–(35).

The statement asks that one polynomial be non-negative wherever two others are. The certificate proves it by an identity that writes a positive multiple of G as $\sigma_0 + \sigma_1 U_h + \sigma_2 Q_h + \sigma_3 U_h Q_h$, where each σ_a is a sum of squares of polynomials. Such a σ_a is non-negative everywhere, so the right-hand side is non-negative wherever U_h and Q_h are. Given such an identity, (40) follows in two lines (Theorem 12.2), and verifying the identity is a computation of the kind the verifier already performs for (H2) and (H3): expand both sides and compare integer coefficients. The method of certifying polynomial inequalities on semialgebraic sets by sums of squares goes back to the Positivstellensätze of real algebraic geometry and to its computational development by Lasserre and Parrilo [12, 24].

12.1. Sums of squares, and how a matrix certifies one. Fix an integer $d \geq 0$ and let $\mathbf{m}_d(q)$ denote the column vector of all $n_d = \binom{d+3}{3}$ monomials $q_0^{e_0} q_1^{e_1} q_2^{e_2} q_3^{e_3}$ of total degree d , listed in a fixed order. The verifier uses the lexicographic order of the exponent vectors.

Lemma 12.1. *Let $\Sigma \in \mathbb{R}^{n_d \times n_d}$ be symmetric positive semidefinite, and define*

$$\sigma(q) = \mathbf{m}_d(q)^\top \Sigma \mathbf{m}_d(q).$$

Then σ is a homogeneous polynomial of degree $2d$ and $\sigma(q) \geq 0$ for every $q \in \mathbb{R}^4$. Moreover σ is a sum of squares of forms of degree d .

Proof. Each entry of $\mathbf{m}_d(q)\mathbf{m}_d(q)^\top$ is a monomial of degree $2d$, so σ is a form of degree $2d$. For fixed q the number $\sigma(q) = \mathbf{v}^\top \Sigma \mathbf{v}$ with $\mathbf{v} = \mathbf{m}_d(q)$ is non-negative because $\Sigma \succeq 0$. For the last statement, write $\Sigma = W^\top W$, for instance from the spectral decomposition. Then $\sigma(q) = \|W\mathbf{m}_d(q)\|^2$ is the sum of the squares of the forms $(W\mathbf{m}_d(q))_i$. $\square$

Only the inequality $\sigma \geq 0$ is used below, and it needs only $\Sigma \succeq 0$. Positive semidefiniteness of a rational Σ is established from a deposited integer witness by Proposition 11.2, exactly as for the matrices C_ℓ .

12.2. The certificate identity.

Theorem 12.2. *Fix a certificate at (τ, L, T^*) , let G be its cover polynomial (38), and let Q_h, U_h be as in (34)–(35). Suppose we are given an integer $k \geq 0$ with $L + k \geq 4$ and four symmetric positive semidefinite matrices with rational entries,*

$$\Sigma_0 \in \mathbb{Q}^{n_{L+k} \times n_{L+k}}, \quad \Sigma_1, \Sigma_2 \in \mathbb{Q}^{n_{L+k-2} \times n_{L+k-2}}, \quad \Sigma_3 \in \mathbb{Q}^{n_{L+k-4} \times n_{L+k-4}},$$

such that, with $\sigma_a = \mathbf{m}^\top \Sigma_a \mathbf{m}$ formed from the monomial vector of the matching degree, the identity

$$(|q|^2)^k G = \sigma_0 + \sigma_1 U_h + \sigma_2 Q_h + \sigma_3 U_h Q_h \quad (41)$$

holds in $\mathbb{Q}[q_0, q_1, q_2, q_3]$. Then hypothesis (H6) holds.

Proof. Let $q \in \mathcal{D}_h(\tau)$, so that $U_h(q) \geq 0$ and $Q_h(q) \geq 0$. By Lemma 12.1 each $\sigma_a(q) \geq 0$, so every one of the four summands on the right-hand side of (41) is non-negative at q , and therefore $(|q|^2)^k G(q) \geq 0$. Since $q \neq 0$, dividing by $(|q|^2)^k > 0$ gives $G(q) \geq 0$. $\square$

12.3. What is deposited, and how it is verified. For each certificate the following records are deposited, in the format of Appendix B:

- the integer k of (41), as `sosExponentK`,
- an even integer κ_s , as `sosDenLog2`,
- the integer symmetric matrices `sosGram` $000a = 2^{\kappa_s} \Sigma_a$ for $a = 0, \dots, 3$,
- for each a , the integer witnesses `sosSwit` $000a$ and `sosWwit` $000a$ of Proposition 11.2.

The verifier then performs the following steps, all in exact arithmetic.

- (S1) *Shape*. Each deposited $2^{\kappa_s} \Sigma_a$ is symmetric and of the size prescribed by Theorem 12.2, namely n_{L+k} , n_{L+k-2} , n_{L+k-4} , computed from the deposited L and k .
- (S2) *Positivity*. For each a , the matrix $E_a = S_a(2^{\kappa_s} \Sigma_a) - W_a^\top W_a$ is symmetric and diagonally dominant, whence $\Sigma_a \succeq 0$ by Proposition 11.2.
- (S3) *Assembly*. The polynomial G is built from the deposited C_ℓ , the universal tables and the weights, as in Section 11.4. The generators Q_h and U_h are read from the certificate records and validated against (34)–(35) at the deposited A and B (check (V2) of Appendix B). The left-hand side of (41) is G multiplied k times by $|q|^2$.
- (S4) *The identity*. The right-hand side of (41) is expanded: for each a and each pair $i \leq j$ of indices, the entry $(\Sigma_a)_{ij}$ contributes $(2 - \delta_{ij})(\Sigma_a)_{ij}$ times the monomial $(m)_i(m)_j$, and the blocks $a = 1, 2, 3$ are then multiplied by U_h , Q_h and $U_h Q_h$ respectively. Both sides are scaled by a common denominator 2^{κ_s} . The identity becomes an equality between two lists of n_{2L+2k} integers, compared entry by entry.

If (S1)–(S4) succeed, then Theorem 12.2 applies and (H6) holds. The number of coefficient equations in (S4) is $n_{2L+2k} = \binom{2L+2k+3}{3}$. All intermediate integers are bounded by an explicit function of the deposited scales and are checked against the machine word width as in Appendix B. A reader who wishes to reimplement the check needs only a parser for the format of Appendix B and exact polynomial multiplication in four variables.

Hypotheses (H1)–(H6) are established for each of the deposited certificates, so Theorem 9.2 yields the bounds of Table 2, and with them the cases $4 \leq n \leq 7$ of Theorem 1.5.

Remark 12.3. The window configurations come from numerical optimization of the minimum pairwise distance, followed by rounding to rational tangent lines. The certificates come from semidefinite programs. Their numerical solutions were rounded to rational data and then repaired so that all identities hold exactly.

13. OPEN PROBLEMS

The smallest open case of Question 1.1 is $n = 8$. We believe that the answer is “no”.

Conjecture 13.1. $(8, \omega_8) \in \mathcal{P}_{\text{cyl}}$.

Question 1.2 is open for every $n \geq 5$.

Problem 13.2. Determine R_n for $n \geq 5$ [4, Section 2.4, Problem 17].

Brass, Moser, and Pach [4, Section 2.4] also asked for the asymptotic behavior of R_n . The ring gives $\liminf_{n \rightarrow \infty} nR_n \geq \pi$.

Problem 13.3. Determine $\lim_{n \rightarrow \infty} nR_n$, or show that it does not exist.

APPENDIX A. ELEMENTARY ESTIMATES

This appendix proves the estimates (E1)–(E4) of Section 5, used in Lemmas 5.1, 5.2 and 6.4.

Lemma A.1. Statements (E1)–(E4) hold.

Proof. (E1): $f(x) = \sin x - x + \frac{x^3}{6}$ has $f(0) = 0$ and $f'(x) = \cos x - 1 + \frac{x^2}{2} \geq 0$, since $g(x) = \cos x - 1 + \frac{x^2}{2}$ has $g(0) = 0$ and $g'(x) = x - \sin x \geq 0$. The upper bound is standard.

(E2): $\frac{\tan y}{y}$ is increasing on $(0, \frac{\pi}{2})$ because $(y \sec^2 y - \tan y) = \frac{y - \sin y \cos y}{\cos^2 y} > 0$. Hence for $y \in (0, 1]$, $\tan y \leq y \tan 1 < y\sqrt{3}$, using $1 < \frac{\pi}{3}$ and the monotonicity of $\tan$. Then $h(y) = y + y^3 - \tan y$ has $h(0) = 0$ and $h'(y) = 3y^2 - \tan^2 y > 3y^2 - 3y^2 = 0$.

(E3): apply $\tan$ to both sides for the identity. The condition $pq > -1$ keeps the difference in $(-\frac{\pi}{2}, \frac{\pi}{2})$. The inequality $\arctan z \leq z$ holds since $\tan$ is convex on $[0, \frac{\pi}{2})$.

(E4): the derivative has numerator $1 + S^2 T^2 + z^2 > 0$.

□

APPENDIX B. DEPOSITED DATA AND STANDALONE VERIFIER

The data are in plain-text files in a self-describing format, designed to be read by a short standalone parser. Each record consists of a key, written `[key]Name[/key]`, followed by a value block `[value] ... [/value]`. A value block holds one or more rows. Each row, and each entry within a row, is delimited by `{` and `}`. Every value is therefore a nested list of integers of depth two, including those that carry a single number, which appear as one row with one entry. Keys that depend on an index ℓ or a end in it as a four-digit number.

The *universal* records, identical in every certificate file, are, for each $\ell = 1, \dots, L$:

- `WNum0000 $\ell$` and `WDen0000 $\ell$` : the weights w_i^ℓ as numerator/denominator pairs, forming

$$D_\ell = \text{diag}(w_1^\ell, \dots, w_{2\ell+1}^\ell);$$

- `Tpoly0000 $\ell$` together with the scale `Tden0000 $\ell$` . Each row has 7 numbers $i, j, e_0, e_1, e_2, e_3, c$. These seven numbers mean that the polynomial $T_h^\ell(q)_{ij}$ contains the term $(c/T\text{den}_\ell) q_0^{e_0} q_1^{e_1} q_2^{e_2} q_3^{e_3}$. The polynomial $T_h^\ell(q)_{ij}$ is the sum over the rows carrying that i, j .

Each *certificate* file records:

- `Lmax`: the truncation L . All degree-dependent constants of the verifier are derived from it;
- `targetBound`: the positive integer N of the claim. It enters the verification only through the counting check (V6);
- `tauSqNum` and `tauSqDen`: the positive integers A and B with $\tau^2 = A/B$;
- `UhPoly` and `QhPoly`: the integer coefficient lists of the generators U_h and Q_h , in the row format of `Tpoly`. The verifier also constructs these polynomials from the closed forms (34)–(35) at the deposited A and B , and checks that the deposited lists agree with them (check (V2) below);
- `MNum`, `MDen`, `TstarNum`, `TstarDen`: a rational number $M < 0$ and the trace sum T^* . The files are not normalized. Dividing the matrices C_ℓ , the number T^* , the polynomial G and the matrices Σ_a by $-M$ gives a certificate in the sense of Definition 9.1. All checks below are invariant under this scaling;
- `cDenLog2`: the even integer $\kappa \in [0, 60]$ scaling the next record;
- `Cscaled0000 $\ell$` = $2^\kappa C_\ell$, as the rows of an integer matrix;
- `Swit0000 $\ell$` and `Wwit0000 $\ell$` : the integer scalar S_ℓ and the integer matrix W_ℓ of Proposition 11.2;
- `sosExponentK`: the integer k of the identity (41);
- `sosDenLog2`: the even integer κ_s scaling the next record;
- `sosGram0000 $a$` = $2^{\kappa_s} \Sigma_a$ for $a = 0, \dots, 3$, as the rows of integer symmetric matrices of the sizes prescribed by Theorem 12.2;
- `sosSwit0000 $a$` and `sosWwit0000 $a$` : the integer scalar S_a and the integer matrix W_a of Proposition 11.2 for Σ_a .

Every other quantity in the proof is derived from these records: the reciprocals $1/w_i^\ell$ from `WNum`/`WDen`, and the polynomials P and $M(|q|^2)^L - P$ from the C_ℓ , the tables and the weights. Apart from U_h , Q_h and the powers of $|q|^2$, the verifier constructs no polynomial in $q_0, \dots, q_3$ other than by expanding deposited records.

On these records the standalone verifier confirms, for each certificate in turn:

- (V1) the claim is well formed: `targetBound` ≥ 1 , $M < 0$, and $0 < \tau^2 < 4$, so that $\omega = \tau/(2 - \tau)$ is defined and positive;
- (V2) the deposited generator records `UhPoly` and `QhPoly` coincide, coefficient by coefficient, with the closed forms (34)–(35) evaluated at the deposited A and B ;
- (V3) (H1) and the positivity of the weights, by inspection of the exponents in `Tpoly0000 $\ell$` and of the signs in `WNum0000 $\ell$` , `WDen0000 $\ell$` ; (H2) and (H3), by expanding both sides of the two identities of Section 11.1 and comparing them coefficient by coefficient (once, since the tables are universal);
- (V4) the hypothesis of Proposition 11.2, forming $E_\ell = S_\ell(2^\kappa C_\ell) - W_\ell^\top W_\ell$ and testing symmetry and diagonal dominance;
- (V5) (H5), by evaluating $\sum_\ell \sum_i (C_\ell)_{ii}/w_i^\ell$ and comparing it with the deposited T^* ;
- (V6) the *counting inequality*

$$T^* < (-M) \cdot \text{targetBound}, \quad \text{equivalently} \quad \text{targetBound} \geq 1 + \left\lceil \frac{T^*}{-M} \right\rceil,$$

in exact rational arithmetic, which by Theorem 9.2 turns hypotheses (H1)–(H6) into the bound $N_{\max}(\tau) \leq \text{targetBound}$;

- (V7) the hypothesis of Proposition 11.2 for each multiplier matrix Σ_a , from the records `sosSwit0000 $a$` and `sosWwit0000 $a$` , establishing $\Sigma_a \succeq 0$ (step (S2) of Section 12);

(V8) (H6), by expanding both sides of the identity (41) and comparing them coefficient by coefficient (steps (S1), (S3) and (S4) of Section 12). The conclusion follows by Theorem 12.2.

Every one of these checks is exact. The verifier uses no floating-point arithmetic. Integers are held in fixed-width machine words of 128, 256 or 512 bits, selected from the deposited L . Each operation is checked for overflow. For rational numbers, we use two representations. If the number is not dyadic, then we store the numerator and denominator. In the dyadic case the denominator is a power of two, and we keep only its exponent. Addition, multiplication and comparison are implemented so that no rounding can occur, and every operation is guarded by an overflow test. Any guard that fires sets a permanent flag. A flagged run reports the fact and its output is discarded.

The verifier’s contract. The verifier is a single program with a one-line verdict. Invoked without arguments it prints its own contract, which we reproduce in substance:

The input file contains, among other data, a positive integer `targetBound` and positive integers `tauSqNum`, `tauSqDen` with `tauSqNum < 4 · tauSqDen`. Define $\tau = \sqrt{\text{tauSqNum}/\text{tauSqDen}}$ and $\omega = \tau/(2 - \tau)$. If the last line printed by the program is THE THEOREM WAS PROVED, then the following statement is a theorem: *no more than targetBound pairwise non-overlapping cylinders of radius ω can simultaneously touch the unit ball*. Everything else in the input file is a machine-generated witness. No modification of the input, however adversarial, can make the program print the verdict for a false statement. The only realistic effect of tampering is that the proof is not found.

In the language of Definition 1.3: the verdict certifies $(\text{targetBound}, \omega) \in \mathcal{P}_{\text{cyl}}$.

The final line is printed only if all of (V1)–(V8) succeed and every arithmetic flag is still clear at the end of the run. Any failure, at any stage, suppresses it. The soundness of the verdict therefore rests on the verifier source, the compiler and hardware executing it, and Theorem 9.2. Two input switches, `skipMultVerification` and `skipOrthVerification`, allow the universal checks in (V3) to be skipped when the tables have been verified separately. If either switch is active, the verdict is conditional on those identities. Both switches are off in all four deposited certificates.

The window configurations of Section 6 are deposited in the format described there, and are checked together with the inequalities of Table 1 by the run `cylie -i constructions` of Section 5.4.

The rational configuration of Remark 3.6 is deposited in `cyl_window_E5.txt`. The file records five rows and a rational number $\tau = \frac{13600}{11513}$. The same run checks that the rows are rational tangent lines with pairwise distances at least τ . Moreover $5 - 2\tau^2 > 0$ and $(5 - 2\tau^2)^2 < 5$, so $\tau^2 > \frac{5-\sqrt{5}}{2} = 4 \sin^2 \frac{\pi}{5}$. Hence the five lines give five non-overlapping cylinders of radius $\frac{\tau}{2-\tau} > \omega_5$.

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ON THE USE OF AI

Ideas were discussed and the literature was reviewed with the help of Anthropic’s Claude models. The programs that searched for certificates and for the configurations of Section 6 were written with the assistance of these models. The verifier [14] was written by the authors and reviewed and tested both by the authors and by the models. The proofs use the certificates and configurations only through the verifier, and do not depend on how they were found. The text was written by the authors and then proofread and edited with the same models.

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